ADVANTEST CORPORATION

V93000 Application Note

The Wave Scale MX Handbook

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0. Document History

Date	Description	Author
04/16/15	Initial Draft	Stefan Groß
10/12/17	Release 8.1.0	Stefan Groß
10/15/17	Examples using DS-API added	Stefan Groß

1. The Wave Scale MX family of instruments for the V93000 test system

Wave Scale MX (Mixed Signal) boards for the V93000 test system come in three different flavors:

- ***E9740HS*** high speed board:
  16 high speed AWG and 16 high speed digitizers
- ***E9740HR*** high resolution board:
  16 high resolution AWG and 16 high resolution digitizers
- ***E9740A*** mixed high speed and high resolution board:
  8 high resolution AWG, 8 high resolution digitizers, 8 high speed AWG and 8 high speed digitizers

All channels are independent and fully controlled by the test processor. The high speed board has an integrated signal processing unit (SPU) for filtering, decimation and down-conversion. The memory is pooled.

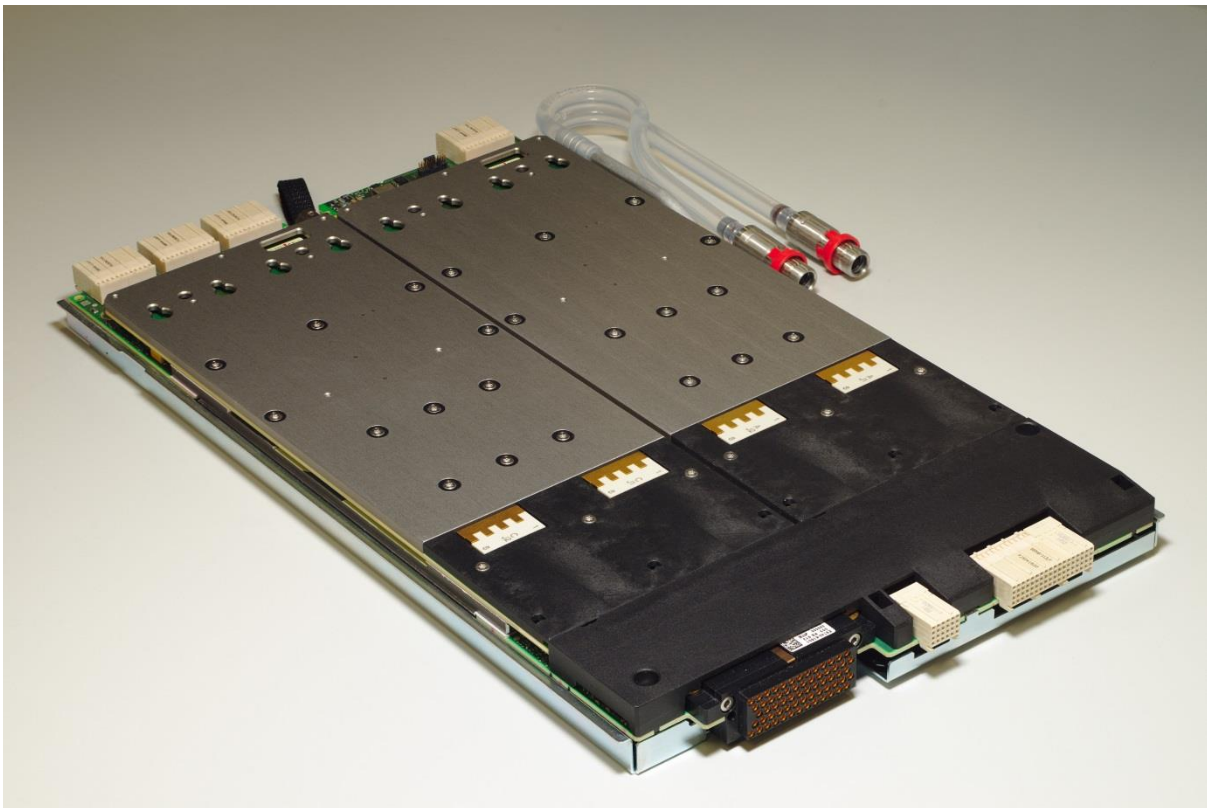

Figure 1: WSMX HS board

There are 16 source and 16 measure units with two bidirectional, differential pogo pairs on each Wave Scale MX E9740 board. Each unit consists of an AWG and a digitizer.

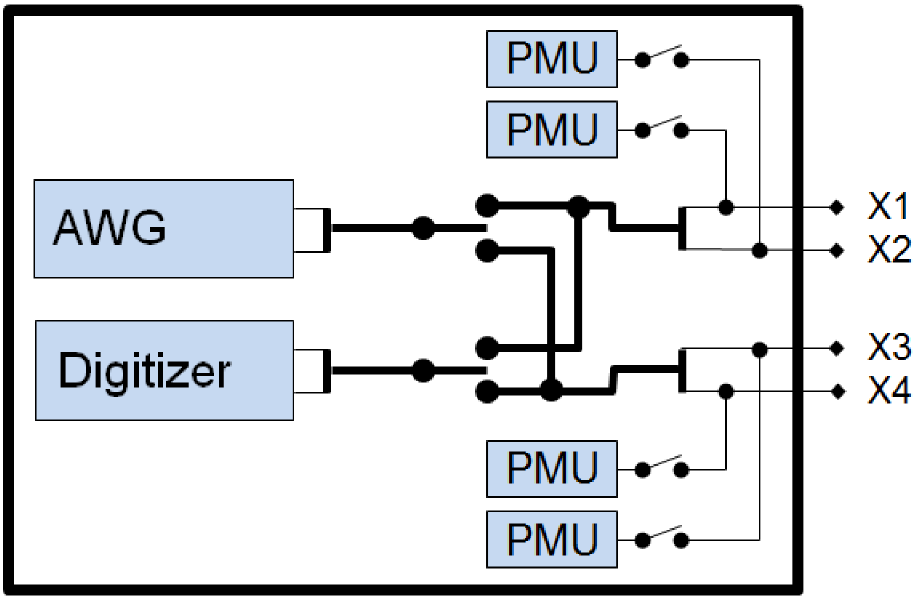

Figure 2: WSMX instrument unit

The AWG and the digitizer of a unit operate independently and fully parallel. The AWG and the digitizer can be connected to each pair of pogo pins. It is recommended to route the AWG to the connectors X1 and X2. A PMU is associated with each pogo. With the built-in 2:1 multiplexer the digitizer can be connected to either connector X3 or X4. Consecutive units form an IQ pair (unit 1 with unit 2, unit 3 with unit 4, etc.)

Wave Scale MX high resolution modules cannot connect the digitizer to the X1/X2 pogo pair and the X3/X4 pogo to the AWG at the same time.

Wave Scale MX high speed channels can be paired, to form a complex I/Q pair.

2. Specification

2.1 Wave Scale MX High-Resolution (HR) AWG

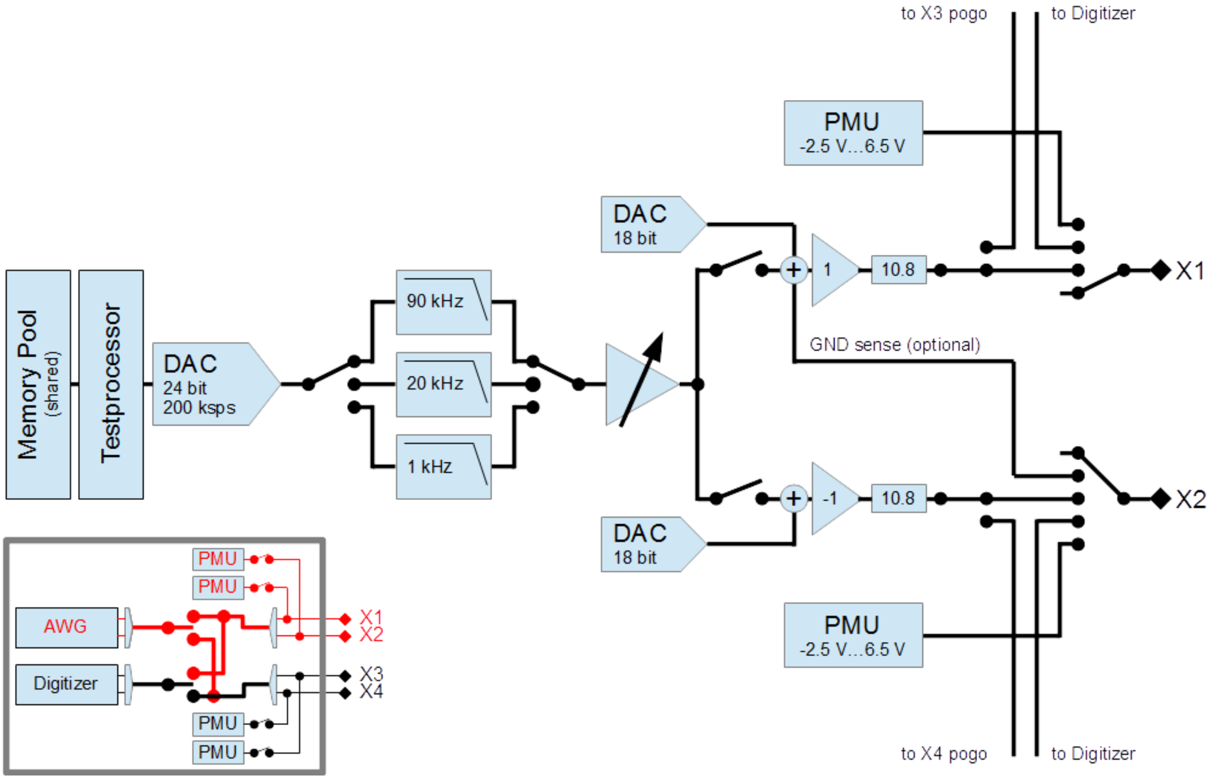

Figure 3: Block diagram of the WSMX HR AWG

Specification	Value
Waveform Memory	144 Msamples shared between 2 AWGs and 2 Digitizers
Sample Rate	10 ksps to 200 ksps
Resolution	24 bit
Max. sinewave frequency	90 kHz
Signal Range	10.3 Vp @ 600 Ω to GND
DC offset range	-10.3 V to 10.3 V
Output level (AC+DC)	-10.3 V < AC+DC < 10.3 V
Output Impedance	X1, X2 pogos: 11.2 Ω ± 5% X3, X4 pogos: 17 Ω ± 30%
Selectable hardware filters	1 kHz, 20 kHz, 90 kHz passband frequency
Max. output current	20 mA
INL (characteristic data)	± 3.5 ppm
DNL (characteristic data)	± 1.5 ppm
Noise Density (typical data)	132 nV/sqrt(Hz) @ 500 Hz, 10 Vp amplitude setting using 1kHz filter 29 nV/sqrt(Hz) @ 20 kHz, 10 Vp amplitude setting using 1kHz filter 18 nV/sqrt(Hz) @ 500 Hz, 0.75 Vp amplitude setting using 1kHz filter 15 nV/sqrt(Hz) @ 20 kHz, 0.75 Vp amplitude setting using 1kHz filter
Short-term drift 0.1 - 10 Hz (characteristic data, differential)	2.2 µVpp, 0.75 Vp amplitude setting using 1kHz filter 26 µVpp, 10 Vp amplitude setting using 1kHz Filter
RMS noise 22 Hz - 22 kHz (typical data, differential)	2.4 µVrms, 0.75 Vp amplitude setting using 1kHz filter 7.3 µVrms, 10 Vp amplitude setting. using 1kHz filter
THD (up to 5th harmonic) 200 kΩ load, differential, THD optimized mode (typical data)	-130 dBc @ 1 kHz, 2Vp -120 dBc @ 1 kHz, 20Vp -117 dBc @ 5 kHz, 2Vp -107 dBc @ 5 kHz, 20Vp -121 dBc @ 20 kHz, 2Vp -104 dBc @ 20 kHz, 20Vp
SFDR [1)] 200 kΩ load, differential (typical data)	126 dBc @ 1 kHz, 2Vp, 20 kHz bw 121 dBc @ 1 kHz, 20Vp, 20 kHz bw 118 dBc @ 5 kHz, 2Vp, 50 kHz bw 108 dBc @ 5 kHz, 20Vp, 50 kHz bw 123 dBc @ 20 kHz, 2Vp, 130 kHz bw 105 dBc @ 20 kHz, 20Vp, 130 kHz bw
SNR 200 kΩ load, differential (typical data)	114 dBc @ 1 kHz, 2Vp, 20 kHz bw [2)] 128 dBc @ 1 kHz, 20Vp 20 kHz bw [2)] 107 dBc @ 5 kHz, 2Vp 50 kHz bw 114 dB @ 5 kHz, 20Vp 50 kHz bw 104 dB @ 20 kHz, 2Vp 130 kHz bw 114 dB @ 20 kHz, 20Vp 130 kHz bw

Table 1: Specifications of the WSMX HR AWG

[1)] SFDR specifies the available dynamic range as the difference in magnitude between the amplitude of the fundamental and the amplitude of the largest spurious component including harmonics in the specified frequency range

[2)] EIAJ A-weighted

2.2 Wave Scale MX High-Resolution (HR) Digitizer

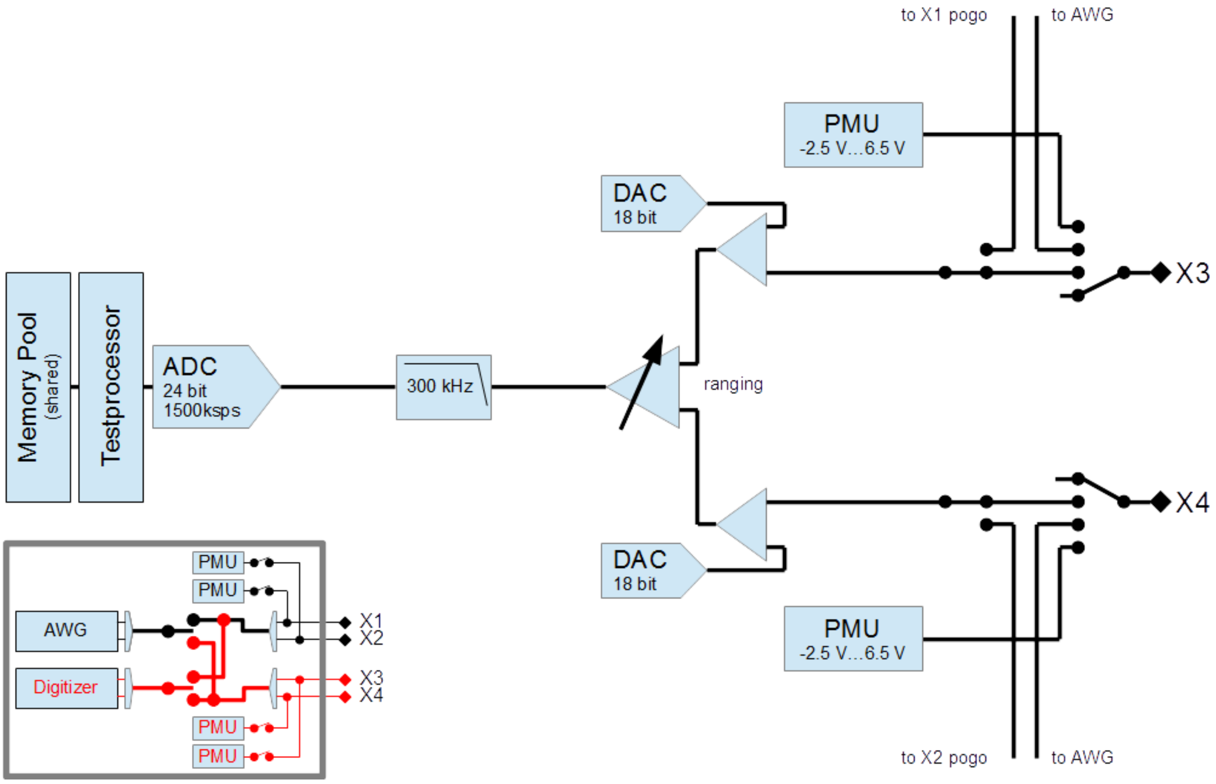

Figure 4: Block diagram of the WSMX HR digitizer

Specification	Value
Waveform Memory	144 Msamples shared between 2 AWGs and 2 Digitizers
Sample Rate	1 ksps to 1500 ksps
Resolution	24 bit
Maximum bandwidth	300 kHz [1)]
Input ranges	10.4 Vp, 5.2 Vp, 2.6 Vp, 1.3 Vp, 520mVp, 260 mVp
DC offset range	10.4 Vp range: no DC offset 5.2 Vp range: no DC offset 2.6 Vp range: -5.2V to +5.2 V 1.3 Vp range: -5.2V to +5.2 V 520 mVp range: -5.2V to +5.2 V 260 mVp range: -5.2V to +5.2 V
Input level (AC+DC)	-10.1 V < AC+DC < 10.1 V $-V_{range} + V_{offset} \leq V_{diff} \leq + V_{range} + V_{offset}$ $-5.2\ V + V_{offset} \leq V_p \leq + V_{offset} + 5.2\ V$ $-5.2\ V - V_{offset} \leq V_n \leq - V_{offset} + 5.2\ V$ $-V_{range} + V_{offset} \leq V_{sig} \leq + V_{range} + V_{offset}$
Input bias current (characteristic data)	± 5 nA
INL (characteristic data)	10.4 Vp range: 2.0 ppm (1.0 ppm with 80 % of range) 5.2 Vp range: 1.6 ppm (0.7 ppm with 80 % of range) 2.6 Vp range: 1.2 ppm (0.7 ppm with 80 % of range) 1.3 Vp range: 1.0 ppm (0.7 ppm with 80 % of range) 520 mVp range: 1.0 ppm (0.8 ppm with 80 % of range) 260 mVp range: 2.5 ppm (1.2 ppm with 80 % of range)
Noise Density w/o DC offset, differential (typical data)	10.4Vp range:144 nV/sqrt(Hz) @ 500Hz, 157 nV/sqrt(Hz) @ 20kHz 5.2Vp range: 81 nV/sqrt(Hz) @ 500Hz, 82 nV/sqrt(Hz) @ 20kHz 2.6Vp range: 36 nV/sqrt(Hz) @ 500Hz, 39 nV/sqrt(Hz) @ 20kHz 1.3Vp range: 20 nV/sqrt(Hz) @ 500Hz, 22 nV/sqrt(Hz) @ 20kHz 520mVp range: 9.8 nV/sqrt(Hz) @ 500Hz, 10.5 nV/sqrt(Hz) @ 20kHz 260mVp range: 7.8 nV/sqrt(Hz) @ 500Hz, 8.2 nV/sqrt(Hz) @ 20kHz
Short-term drift 0.1 - 10 Hz peak-to-peak, differential (characteristic data)	10.4 Vp range: 6.6 µVpp 5.2 Vp range: 3.9 µVpp 2.6 Vp range: 1.7 µVpp 1.3 Vp range: 1.1 µVpp 520 mVp range: 670 nVpp 260 mVp range: 620 nVpp
RMS noise 22 Hz - 22 kHz, w/o DC offset, differential (typical data)	10.4 Vp range: 21.3 µVrms 5.2 Vp range: 12.6 µVrms
THD (up to 5th harmonic) [2)] w/o DC offset, differential (typical data)	-126 dBc @ 1 kHz, 2 Vp -127 dBc @ 1 kHz, 10 Vp -116 dBc @ 5 kHz, 2 Vp -119 dBc @ 5 kHz, 10 Vp -104 dBc @ 20 kHz, 2 Vp -105 dBc @ 20 kHz, 10 Vp

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SFDR[3)] **w/o DC offset, differential (typical data)**	116 dBc @ 1 kHz, 2Vp, 20 kHz bw 118 dBc @ 1 kHz, 10 Vp, 20kHz bw 117 dBc @ 5 kHz, 2 Vp, 50kHz bw 118 dBc @ 5 kHz, 10 Vp, 50kHz bw 107 dBc @ 20 kHz, 2 Vp, 300kHz bw 107 dBc @ 20 kHz, 10 Vp, 300kHz bw
SNR **w/o DC offset, differential (typical data)**	110 dB @ 1 kHz, 2 Vpp 20 kHz bw [2) 4)] 119 dB @ 1 kHz, 2 Vpp 20 kHz bw [2) 5)] 112 dB @ 1 kHz, 10 Vpp 20 kHz bw [2) 4)] 133 dB @ 1 kHz, 10 Vpp 20 kHz bw [2) 5)] 105 dB @ 5 kHz, 2 Vpp 50 kHz bw [4)] 116 dB @ 5 kHz, 2 Vpp 50 kHz bw [5)] 107 dB @ 5 kHz, 10 Vpp 50 kHz bw [4)] 130 dB @ 5 kHz, 10 Vpp 50 kHz bw [5)] 98 dB @ 20 kHz, 2 Vpp 300 kHz bw [4)] 109 dB @ 20 kHz, 2 Vpp 300 kHz bw [5)] 100 dB @ 20 kHz, 10 Vpp 300 kHz bw [4)] 123 dB @ 20 kHz, 10 Vpp 300 kHz bw [5)]

Table 2: Specifications of the WSMX HR digitizer

[1)] This bandwidth is only usable for signal levels up to -9 dBFS in each range. For full-scale signals, the maximum bandwidth is 100 kHz (max. level at 200 kHz: -6 dBFS)
[2)] EIAJ A-weighted
[3)] SFDR specifies the available dynamic range as the difference in magnitude between the amplitude of the fundamental and the amplitude of the largest spurious component including harmonics in the specified frequency range
[4)] SNR is measured with a -60dB signal in the Wave Scale MX digitizer range that fits to the full-scale signal and then referred to the full-scale signal. This measurement technique ensures that the distortion components do not affect the noise measurement and is proposed by the Audio Engineering Society, AES17-1998.
[5)] SNR is measured with a -60dB signal in the smallest possible Wave Scale MX digitizer range and then referred to the full-scale signal. This measurement technique ensures that the distortion components do not affect the noise measurement and is proposed by the Audio Engineering Society, AES17-1998.

2.3 Wave Scale MX High-Speed (HS) AWG

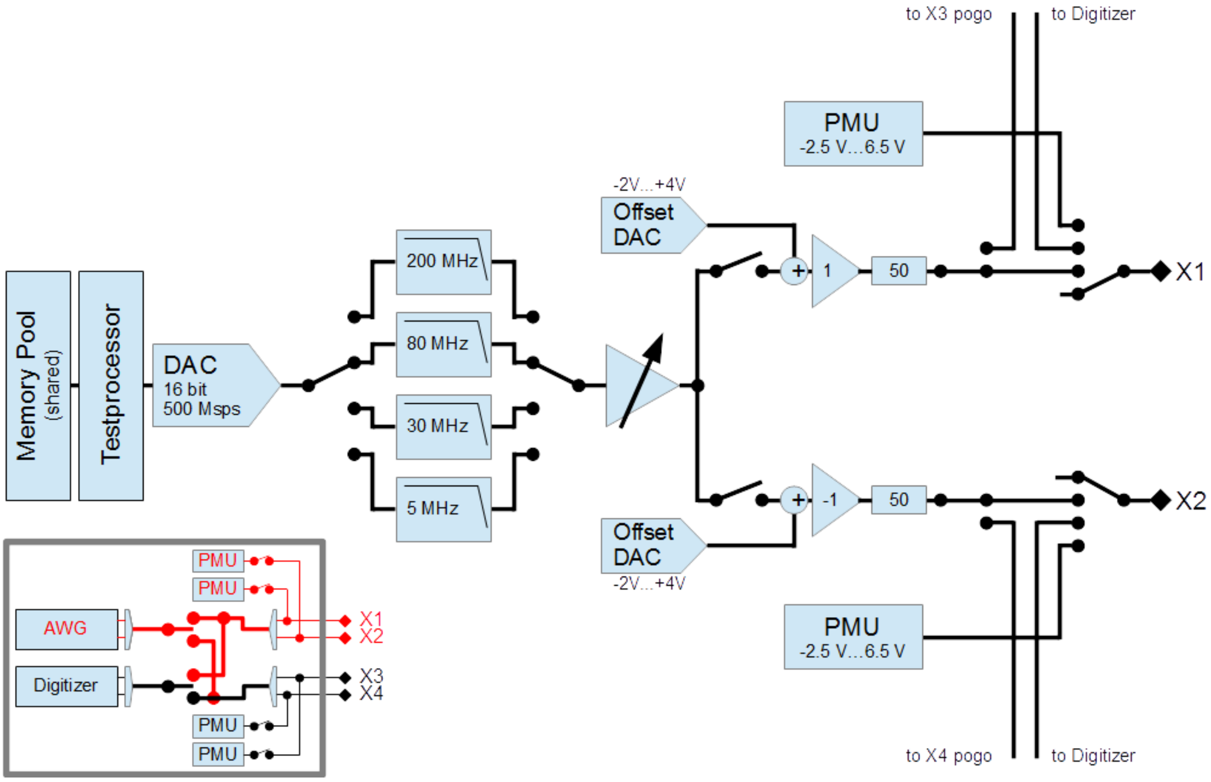


Figure 5: Block diagram of the WSMX HS AWG

Specification	Value
Memory	216 Msamples shared between 2 AWGs and 2 Digitizers
Sample Rate [1)]	10 Msps to 500 Msps
Resolution	16 bit
Max. sinewave frequency	220 MHz (-3dB attenuation frequency)
Amplitude range	80 mVpp to 5 Vpp single-ended into open 40 mVpp to 2.5 Vpp single-ended into 50 Ω additional attenuation is handled in the system software by downscaling of the waveform
DC offset range	-2 V to 4 V (into open)
Output level (AC+DC)	-2 V < AC+DC < 4 V (into open)
Absolute DC accuracy	± (0.8% of setting +8mV +0.2% of DC offset) (into open)
Output DC impedance	50 Ω ± 5%
Analog reconstruction filters (0.1dB attenuation frequency)	5 MHz (-3dB @ 5.7MHz) 30 MHz (-3dB @ 37MHz) 80 MHz (-3dB @ 88MHz) 200 MHz (-3dB @ 220MHz)
Noise density @ 10 MHz (typical data)	-141 dBm/Hz for signal amplitudes <1.75Vpp into open
THD (up to 5th harmonic) [2)] 50 Ω AC termination, 0 dBm Signal, 1 V DC offset, single-ended (typical data) 50 Ω AC termination, -10 dBm Signal, 1 V DC offset, single-ended (typical data)	 -82 dBc @ 5 MHz (5 MHz filter) -81 dBc @ 10 MHz (30 MHz filter) -74 dBc @ 20 MHz (30 MHz filter) -66 dBc @ 50 MHz (80 MHz filter) -55 dBc @ 100 MHz (200 MHz filter) -92 dBc @ 5 MHz (5 MHz filter) -85 dBc @ 10 MHz (30 MHz filter) -84 dBc @ 20 MHz (30 MHz filter) -75 dBc @ 50 MHz (80 MHz filter) -61 dBc @ 100 MHz (200 MHz filter)
SFDR [2)3)] 50 Ω AC termination, 0 dBm Signal, 1 V DC offset, single-ended (typical data) 50 Ω AC termination, -10 dBm Signal, 1 V DC offset, single-ended (typical data)	 81 dB @ 5 MHz (5 MHz filter, 25 MHz bw) 81 dB @ 10 MHz (30 MHz filter, 50 MHz bw) 77 dB @ 20 MHz (30 MHz filter, 100 MHz bw) 66 dB @ 50 MHz (80 MHz filter, 250 MHz bw) 56 dB @ 100 MHz (200 MHz filter, 250 MHz bw) 75 dB @ 5 MHz (5 MHz filter, 25 MHz bw) 82 dB @ 10 MHz (30 MHz filter, 50 MHz bw) 79 dB @ 20 MHz (30 MHz filter, 100 MHz bw) 70 dB @ 50 MHz (80 MHz filter, 250 MHz bw) 60 dB @ 100 MHz (200 MHz filter, 250 MHz bw)

SNR 2)	
50 Ω AC termination, 0 dBm Signal, 1 V DC offset, single-ended (typical data)	67 dB @ 5 MHz (5 MHz filter, 25 MHz bw) 64 dB @ 10 MHz (30 MHz filter, 50 MHz bw) 61 dB @ 20 MHz (30 MHz filter, 100 MHz bw) 57 dB @ 50 MHz (80 MHz filter, 250 MHz bw) 54 dB @ 100 MHz (200 MHz filter, 250 MHz bw)
50 Ω AC termination, -10 dBm Signal, 1 V DC offset, single-ended (typical data)	58 dB @ 5 MHz (5 MHz filter, 25 MHz bw) 54 dB @ 10 MHz (30 MHz filter, 50 MHz bw) 51 dB @ 20 MHz (30 MHz filter, 100 MHz bw) 48 dB @ 50 MHz (80 MHz filter, 250 MHz bw) 48 dB @ 100 MHz (200 MHz filter, 250 MHz bw)
SINAD 2)	
50 Ω AC termination, 0 dBm Signal, 1 V DC offset, single-ended (typical data)	67 dB @ 5 MHz (5 MHz filter, 25 MHz bw) 64 dB @ 10 MHz (30 MHz filter, 50 MHz bw) 61 dB @ 20 MHz (30 MHz filter, 100 MHz bw) 55 dB @ 50 MHz (80 MHz filter, 250 MHz bw) 52 dB @ 100 MHz (200 MHz filter, 250 MHz bw)
50 Ω AC termination, -10 dBm Signal, 1 V DC offset, single-ended (typical data)	57 dB @ 5 MHz (5 MHz filter, 25 MHz bw) 54 dB @ 10 MHz (30 MHz filter, 50 MHz bw) 51 dB @ 20 MHz (30 MHz filter, 100 MHz bw) 48 dB @ 50 MHz (80 MHz filter, 250 MHz bw) 48 dB @ 100 MHz (200 MHz filter, 250 MHz bw)
Dual-tone 3rd order intermodulation distortion 50 Ω AC termination, -6 dBm Signal each tone, 1 V DC offset, single-ended (typical data)	-79 dBc @ 1 MHZ + 1.1 MHz -79 dBc @ 10 MHz + 11 MHz -69 dBc @ 75 MHz + 80 MHz

Table 3: Specifications of the WSMX HS AWG

1) Lower sample rates are automatically resampled by SmarTest to a range supported by the hardware.
2) AC characteristics are measured using signals created by the waveform generation algorithm for sinusoidal signals embedded into SmarTest
3) SFDR specifies the available dynamic range as the difference in magnitude between the amplitude of the fundamental and the amplitude of the largest spurious component including harmonics in the specified frequency range

2.4 Wave Scale MX High-Speed (HS) Digitizer

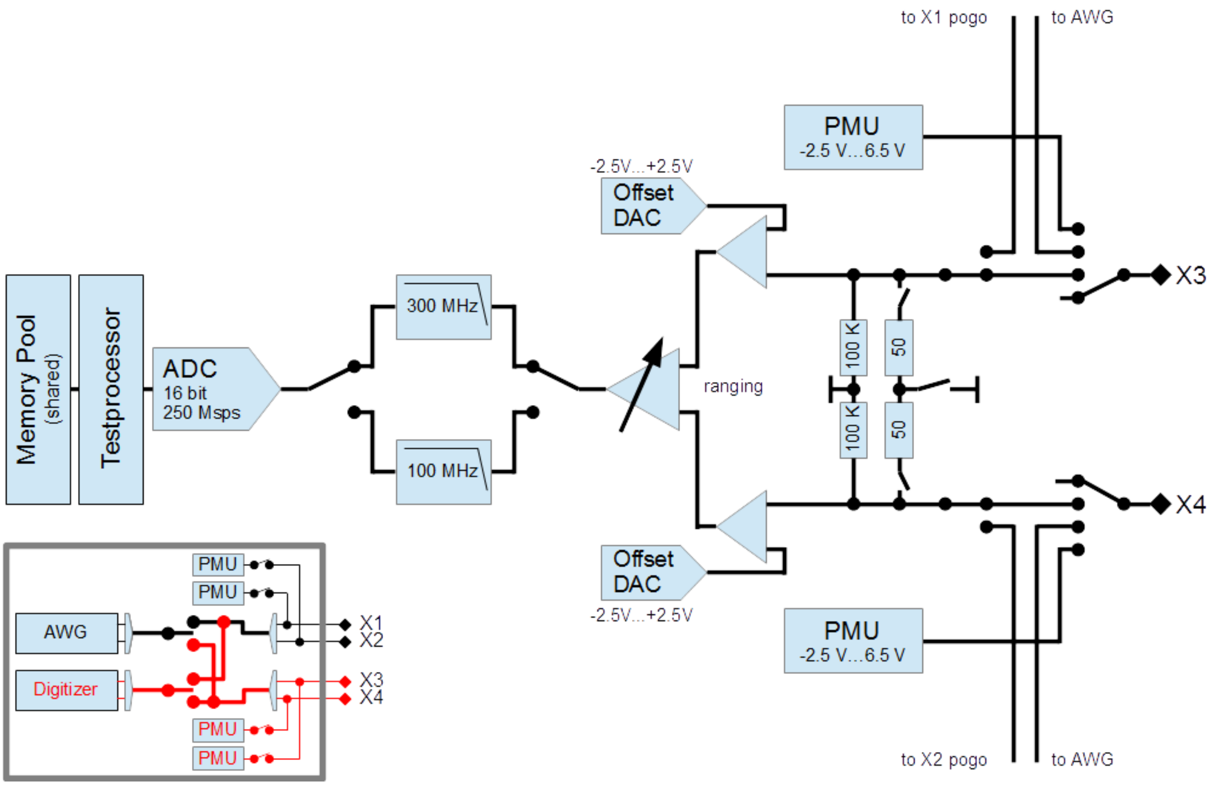
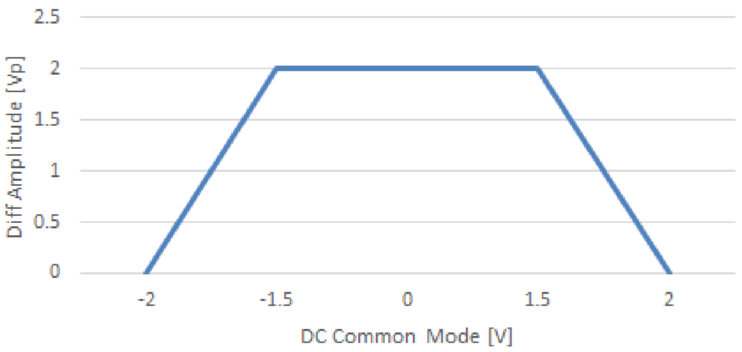


Figure 6: Block diagram of the WSMX HS digitizer

Specification	Value
Memory	216 Msamples shared between 2 AWGs and 2 Digitizers
Sample Rate	6.25 Msps to 250 Msps
Resolution	16 bit
Maximum input bandwidth	300 MHz
DC offset range (single ended)	-2.5 V ... 2.5 V
DC common mode (differential) The combination of DC common mode and the amplitude of a differential signal must remain within the area below the curve otherwise clipping will occur.	
Input range (single ended)	62.5mVpp ... 3Vpp
Compliance Range (maximum external voltage, connected state)	-5V to +5V
Absolute DC accuracy	For signal levels (expectedAmplitude property) < 355mVp: ±(0.5% of reading + 0.75% of offset + 5mV) @ 100kΩ input impedance For signal levels (expectedAmplitude property) > 355mVp: ±(1.125% of reading + 0.75% of offset + 7.5mV) @ 100kΩ input impedance
Input DC impedance	Single-ended: 50Ω ± 5%, 100kΩ ±10% Differential: 100Ω ± 5%, 100kΩ ± 10% to center ground
Analog Filters	100 MHz 300 MHz
Noise density (typical data)	142 dBm/Hz for expected amplitude setting of 100mVp 141 dBm/Hz for expected amplitude setting of 200mVp 139 dBm/Hz for expected amplitude setting of 300mVp 130 dBm/Hz for expected amplitude setting of 500mVp
THD (up to 5th harmonic)	
50 Ω termination, 250 mVp input signal, w/o DC offset, single-ended (typical data)	-79 dBc @ 5 MHz -79 dBc @ 10 MHz -74 dBc @ 50 MHz -66 dBc @ 80 MHz
50 Ω termination, 75 mVp input signal, w/o DC offset, single-ended (typical data)	-81 dBc @ 5 MHz -81 dBc @ 10 MHz -78 dBc @ 50 MHz -71 dBc @ 80 MHz

SFDR [1)]	
50Ω termination, 250mVp input signal w/o DC offset, single-ended (typical data)	80 dBc @ 5 MHz 80 dBc @ 10 MHz 73 dBc @ 50 MHz 68 dBc @ 80 MHz
50Ω termination, 75mVp input signal, w/o DC offset, single-ended (typical data)	79 dBc @ 5 MHz 79 dBc @ 10 MHz 72 dBc @ 50 MHz 68 dBc @ 80 MHz
SNR	
50Ω termination, 250mVp input signal, w/o offset, single-ended (typical data)	63 dB @ 5MHz (15MHz bw, 100MHz filter) 58 dB @ 10MHz (50MHz bw, 100MHz filter) 53 dB @ 50MHz (100MHz bw, 100MHz filter) 49 dB @ 80MHz (300MHz bw, 300MHz filter)
50Ω termination, 75mVp input signal, w/o offset, single-ended (typical data)	55 dB @ 5MHz (15MHz bw, 100MHz filter) 50 dB @ 10MHz (50MHz bw, 100MHz filter) 44 dB @ 50MHz (100MHz bw, 100MHz filter) 41 dB @ 80MHz (300MHz bw, 300MHz filter)
SINAD	
50Ω termination, 250mVp input signal w/o offset, single-ended (typical data)	63 dB @ 5MHz (15MHz bw, 100MHz filter) 58 dB @ 10MHz (50MHz bw, 100MHz filter) 53 dB @ 50MHz (100MHz bw, 100MHz filter) 49 dB @ 80MHz (300MHz bw, 300MHz filter)
50Ω termination, 75mVp input signal w/o offset, single-ended (typical data)	55 dB @ 5MHz (10MHz bw, 100MHz filter) 50 dB @ 10MHz (50MHz bw, 100MHz filter) 44 db @ 50MHz (100MHz bw, 100MHz filter) 41 dB @ 80MHz (300MHz bw, 300MHz filter)
Dual-tone 3rd order intermodulation Distortion 50 Ω termination, 0 dBm Signal, w/o DC offset (characteristic data)	90 dBc @ 1 MHZ + 1.1 MHz 82 dBc @ 10 MHz + 11 MHz

Table 4: Specifications of the WSMX HS digitizer

[1)] SFDR specifies the available dynamic range as the difference in magnitude between the amplitude of the fundamental and the amplitude of the largest spurious component including harmonics in the specified frequency range

3. AWG Setup

The SmarTest8 AWG instrument model is used to set up the WSMX AWG.

3.1 DUT Board

All signals are assigned to pogo numbers in the DUT board file, together with the DUT board properties.

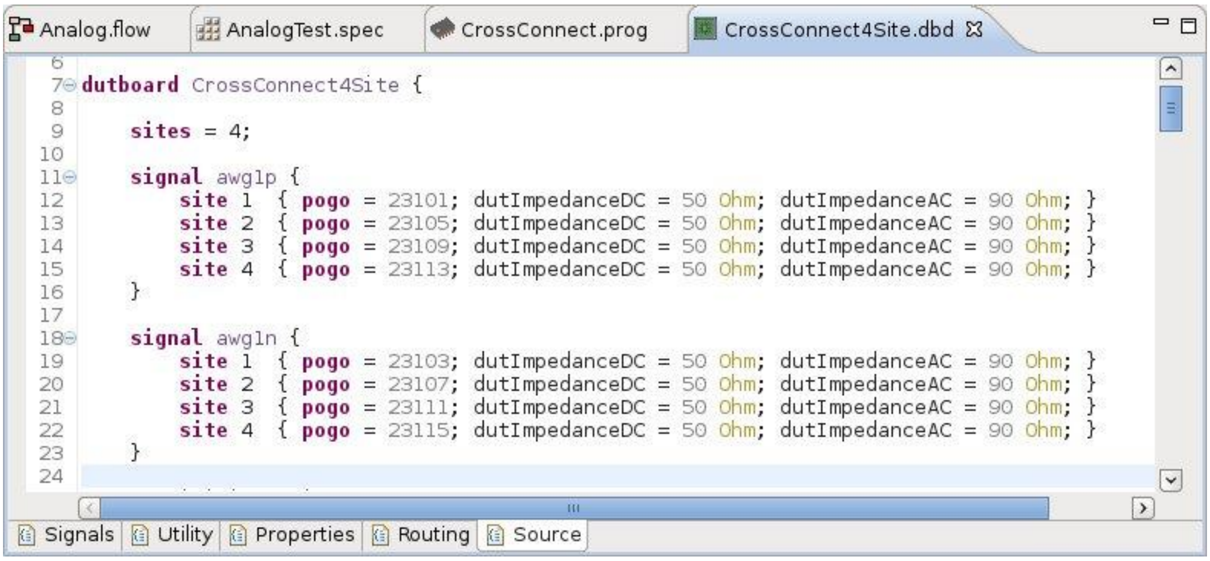
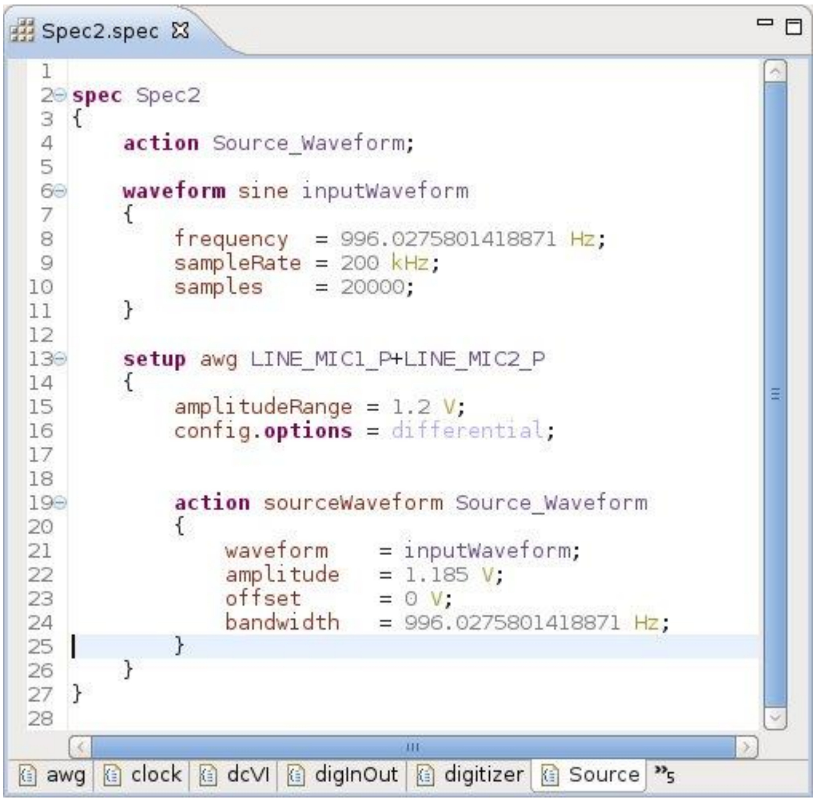


Figure 7: AWG pins in the DUT board file

3.2 AWG instrument properties

Instrument properties apply to all actions of a pin or pin group within one specification and cannot be changed during a measurement run. In the specification file they are located inside the *setup* block.




```
spec Spec2
{
    action Source_Waveform;

    waveform sine inputWaveform
    {
        frequency  = 996.0275801418871 Hz;
        sampleRate = 200 kHz;
        samples    = 20000;
    }

    setup awg LINE_MIC1_P+LINE_MIC2_P
    {
        amplitudeRange = 1.2 V;
        config.options = differential;


        action sourceWaveform Source_Waveform
        {
            waveform    = inputWaveform;
            amplitude   = 1.185 V;
            offset      = 0 V;
            bandwidth   = 996.0275801418871 Hz;
        }
    }
}
```

Figure 8: SmarTest8 specification file

3.2.1 compensation

Use this property to set the quality of the amplitude compensation. Currently only standard compensation is supported.

Example:
```
compensation = standard;
```

3.2.2 connect

By default, an analog instrument is connected right before its first action and only disconnected at the end of the test program. If *connect* is set to *false*, the instrument will be disconnected before the execution and will stay disconnected during the execution. If *connect* is set to *true*, the instrument is connected before the start of the action, and remains connected after the measurement.

Default: *true*

Example:
```
connect = false;
```

3.2.3 disconnect

By default, an analog instrument remains connected once it has been connected, and is only disconnected at the end of the test program. Use this property to disconnect the AWG instrument after the measurement execution by setting *disconnect* to *true*. If set to *false*, the instrument remains connected after the measurement.
Default: *false*

Example:
```
disconnect = true;
```

3.2.4 delay

To fine-tune the synchronization between the AWG and the other instruments the actual start of an action can be delayed with the delay property. Use this property to adjust the phase of the signal or to add a small delay. For larger delays you can use the *wait* statement in the operating sequence or anchor the action to a different vector.
Default: 0 ns

Example:
```
delay = 9.1ns;
```

3.2.5 amplitudeRange

By default SmarTest automatically selects the amplitude range of the AWG based on the amplitude of the stimulus waveform, such that the quantization noise of the AWG is reduced to a minimum. This means that for the vast majority of setups it is not required to set the amplitude range.

For precise amplitude or gain measurements, however, a constant range for all involved AWG actions might be required. In this case a fix amplitude range can be selected, to avoid range changes between actions.
Default: amplitude of the stimulus waveform.

Example:
```
amplitudeRange = 2.1V;
```

3.2.6 superPeriod

This property sets a super period to align all signals in this instrument's phase domain.
Default: LCM of all periods.

Example:
```
superPeriod = 103.1ns;
```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3.2.7 keepAlive

The keep-alive feature provides a continuous, stimulus to the DUT even beyond test suite boundaries. This is crucial for input signals which have to be interrupt-free or in sync with other signals e. g. the clock input of a PLL, which would have to re-synchronize in every test suite otherwise.

If set to *true*, the instrument is locked and keeps its state: If it is sourcing an infinite waveform, it keeps sourcing this waveform. If the waveform is not infinite, it just keeps sourcing the last sample. This means that both r*epeatInfinite* and *keepAlive* have to be set to *true* to keep a periodic waveform running throughout several test suites. While the instrument is locked, it is not available for further actions. The configuration cannot be changed unless *keepAlive* is explicitly cleared.
Default: *false*

Example:

```
keepAlive = true;
```

3.3 AWG instrument options

Instrument properties and instrument options apply to all actions of a pin or pin group within one specification and cannot be changed during a measurement run. In the specification file instrument options are located inside the *setup* block together with the instrument properties.

3.3.1 Differential connection

The property *differential* enables differential connectivity. By default, all connections of analog instruments are single-ended. To connect a Wave Scale MX channel in differential mode, declare the negative leg in the DUT board file and set the property *differential* to *true*. The negative channel does not require a separate setup, but an error is thrown if this signal is not defined in the DUT board file.

Example:

```
config.option = differential;
```

Remark: The DC offset voltage of the positive and negative leg can be set separately. The amplitude always refers to a single channel versus offset, so that in differential mode the resulting differential amplitude is two times this amplitude.

3.3.2 Internal Cross-Connect

Use the property *internalCrossConnect* to internally connect the positive AWG output to the positive digitzer input of the same unit without a load board connection (internal cross-connect). Internal cross-connect is always single-ended. To establish the connection the option must be set for both the AWG and the digitizer.

Example:

```
config.options = internalCrossConnect;
```

3.3.3 Ground sense

In single-ended mode the accuracy of the forced waveform can be improved by sensing the ground plane with the negative leg of the differential pair. This requires a connection between the negative leg and the ground plane on the DUT board. It is recommended to use this option in single-ended setups on Wave Scale high resolution (WSMX HR) modules. This option is only available on Wave Scale high resolution (WSMX HR) modules.

Example:

```
gndSense = true;
```

3.3.4 Resource Sharing

One AWG unit can source a waveform to up to four different sites in single-ended mode, or two sites in differential mode, without external relays. Just assign the four pogo connectors X1, X2, X3 and X4 (single-ended mode) or the two differential pairs X1/X2 and X3/X4 (differential mode) to different sites in the dutboard file.

The two legs of a connector pair (X1/X2 and X3/X4) drive the signal at the same time (fan-out), the two connector pairs are connected sequentially to their respective sites. The phase of the signal on the negative leg is inverted.

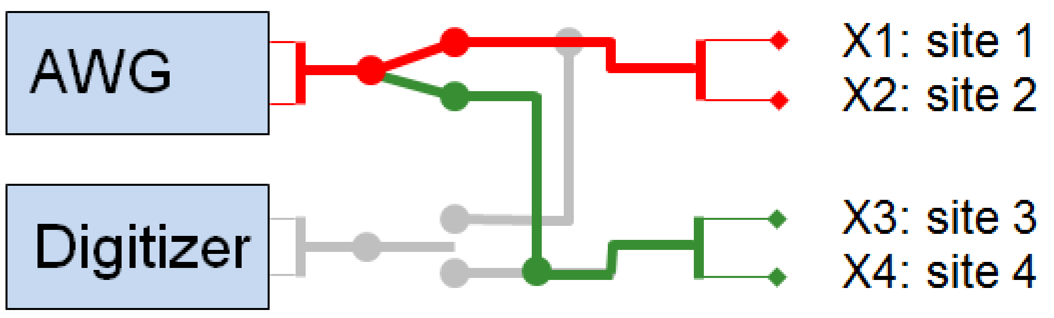


Figure 9: AWG fan out

In the four site setup as shown above, in sequential execution mode, which is the default, two complete measurement runs are executed; one for sites 1 and 2, and one for sites 3 and 4. In interlaced execution mode only one measurement run is executed, and the AWG switches between the sites 1/2 and 3/4 just for the actual measurement.

Example:

```
config.options = siteInterlacingOn;
```

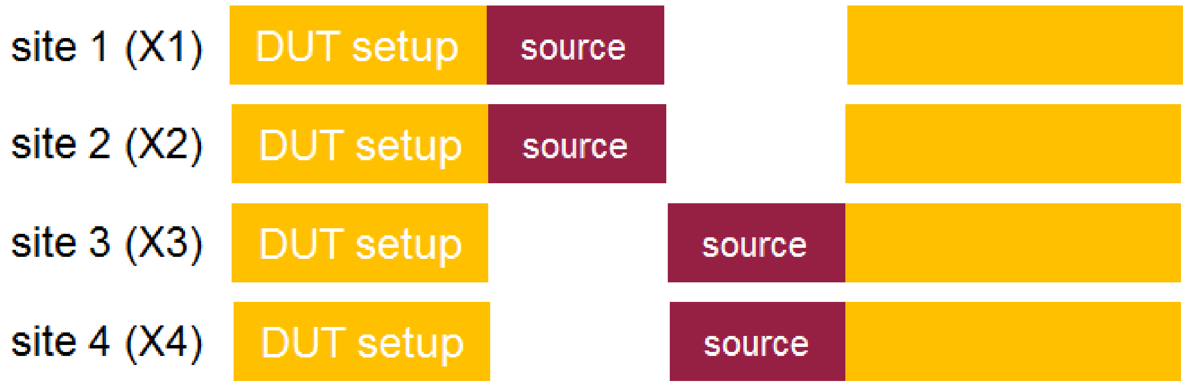


Figure 10: Interlaced execution of an AWG action, four sites

3.4 Analog Waveforms

Analog waveform data used for the waveform definitions in awg, rfStim and dcVI instruments are contained in the waveform specification. They contain the complete signal description including the timing information (sample rate). The waveform description comes in two different flavors:

- *standard waveform* based on a shape (sine, multi-tone, etc.)
- *sample-based waveform* based on a list of individual, equidistant samples

The sample rate is part of the waveform.

WSMX HR

WSMX HR supports AWG sample rates from 10 ksps (10 sps for ramps) to 200 ksps.

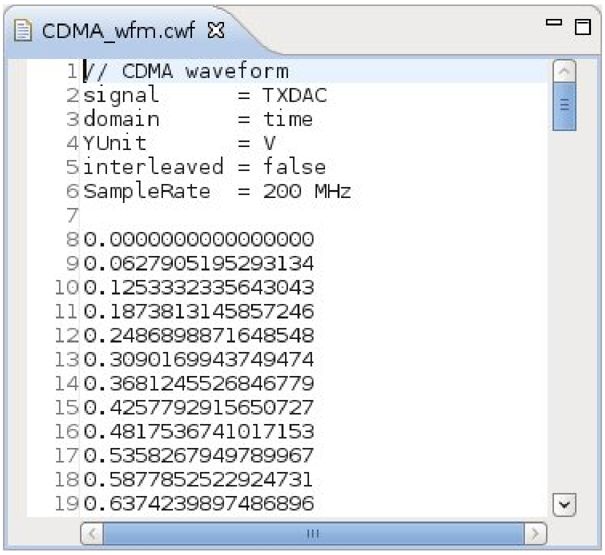
WSMX HS

WSMX HS supports AWG sample rates from 1 Msps to 500 Msps.

3.4.1 Sample-based waveforms

Sample-based waveforms are contained in external files of different formats. The supported file formats are plain ASCII (.cwf), Signal Studio (.wfm) and 89600 (.vsa). Plain ASCII files consist of an optional header section with metadata, and a data section with the actual sample values. The file format is backwards compatible with the custom waveform file format used in SmarTest 7.

The sample rate is either part of the waveform specification or the waveform file header. If both values are set, the value from the specification is taken; resampling is not performed. The format of the numbers within a waveform file is arbitrary.




```
// CDMA waveform
signal      = TXDAC
domain      = time
YUnit       = V
interleaved = false
SampleRate  = 200 MHz

0.0000000000000000
0.0627905195293134
0.1253332335643043
0.1873813145857246
0.2486898871648548
0.3090169943749474
0.3681245526846779
0.4257792915650727
0.4817536741017153
0.5358267949789967
0.5877852522924731
0.6374239897486896
```

Figure 11: Custom waveform file

Please be aware that the DC content of all waveforms sourced by AWGs is removed, and the AC content is rescaled to the amplitude set by the action property *amplitude*. Use the properties *offset* and *offsetNeg* to set the DC offset.

3.4.2 Standard waveforms

Standard waveform descriptions are the preferred way to describe analog waveforms. They are easy to set up, easy to understand and it is easy to modify them. The following standard waveform shapes can be set up:

- *dc*
- *ramp*
- *sine*
- *multi-tone*
- *noise*

The signal quality can differ significantly between setups using standard and sample-based waveforms. The Wave Scale MX high resolution AWG contains different D/A converters, some are optimized for static and others for dynamic signals. Only when standard waveforms are used in the setup, SmarTest selects the most suitable resource, and the instrument setup is optimized for maximum performance (sample rate, filter selection, compensation & pre-distortion).

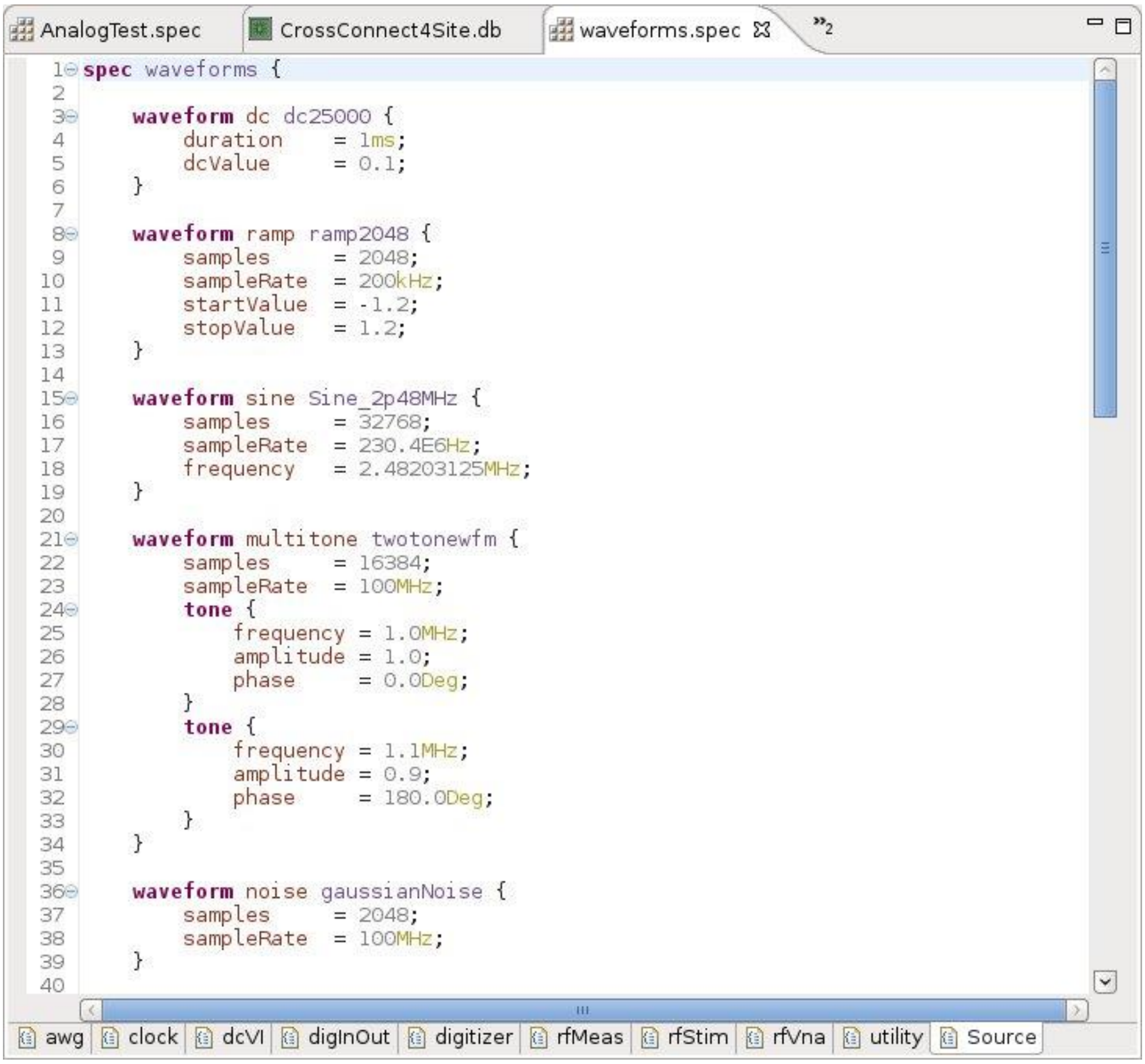


Figure 12: Standard Waveforms

Duration of analog waveforms

Sample size and rate, and the repeat count define the duration of sample-based waveforms. The duration of standard waveforms can also be determined by the property *duration*, and that of sine waves also by the property *periods*.

Set the property *repeatInfinite* to *true* to keep the AWG running until the start of a new parallel block within the operating sequence, the end of the measurement run or even beyond the boundary of the test suite. In the latter case the property *keepAlive* also has to be set to *true.*

If the instrument property *keepAlive* is *true*, the AWG instrument keeps sourcing the waveform even beyond measurement runs and test suite boundaries and stays locked. The only way to turn it off is to execute a separate setup wherein this property is set to *false*.

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3.5 Properties of the *sourceWaveform* action

Action properties are located inside the *action* block of the specification file. They are only effective for one specific action.

3.5.1 Waveform

Use the mandatory property *waveform* to reference the waveform. The I and Q waveforms are contained in separate files. Use the properties *waveformI* and *waveformQ* to reference IQ waveforms.

Examples:
```
waveform = sine1kHz;

waveformI = lte34_I;
waveformQ = lte34_Q;
```

3.5.2 Repeat Count

Use the property *repeatCount* to increase the number of waveform executions. The waveform of a periodic signal can be reduced to a single or a few signal periods, which are then repeated. Use this feature to save waveform memory in the WSMX instrument. Make sure that the waveform contains an exactly integer number of cycles.

Default: 1
This property is mutually exclusive with the property *repeatInfinite*.

Example:
```
repeatCount = 10;
```

3.5.3 Repeat Infinite

The property *repeatInfinite* is useful to adapt the duration of a stimulus automatically to that of a measurement. With this property set to true, the waveform is repeated until a new parallel block is started within the operating sequence, or till the end of the measurement run. The waveform is repeated at least until all other actions or patterns in a sequencer run have finished. This option can slightly increase the duration of a sequencer run.

To keep the waveform running seamlessly throughout several test suites, both *repeatInfinite* and *keepAlive* have to be set to true.

Default: *false*
This property is mutually exclusive with the property *repeatCount.*

Example:
```
repeatInfinite = true;
```

3.5.4 Upsampling

By default, the D/A converter of the AWG samples at the sample rate which is defined in the waveform specification. You can use the option *upsampling* to reduce images by computing additional samples with interpolation techniques. With *upsampling* turned on, the D/A converter runs at a multiple of the original sample rate. This shifts the Nyquist frequency and thus the images to higher frequencies, so that they can be better attenuated by the analog reconstruction filters.

The upsampling option is only available on WSMX HS modules.

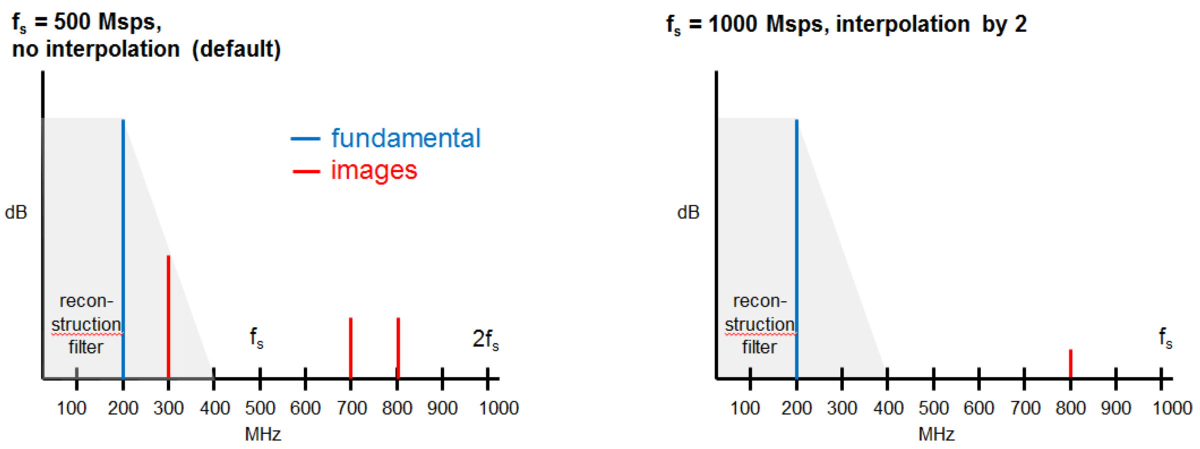


Figure 13: Image reduction through upsampling

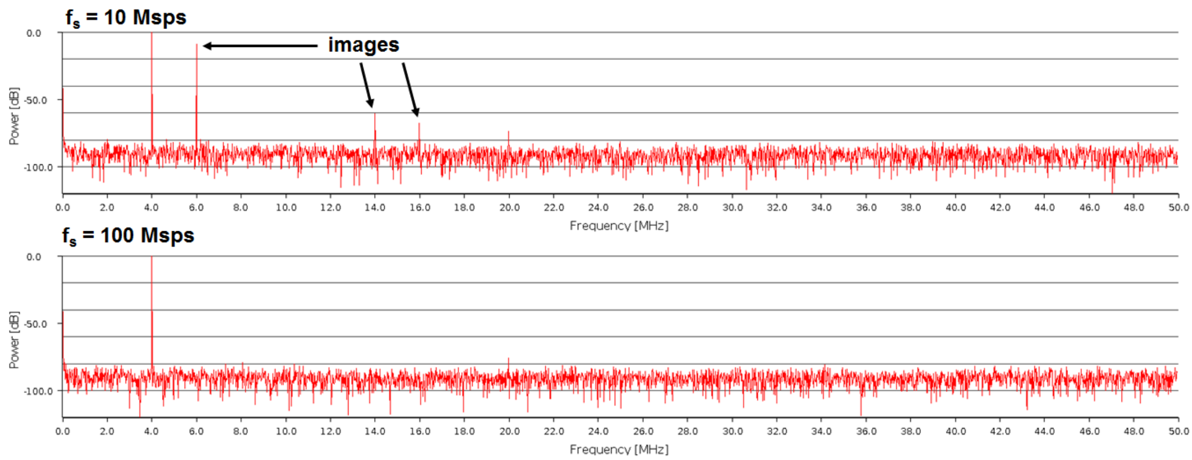


Figure 14: Image reduction of a 4 MHz tone (with 5.7 MHz reconstruction filter)

The WSMX HS hard- and software offers two ways of upsampling: software and hardware upsampling.

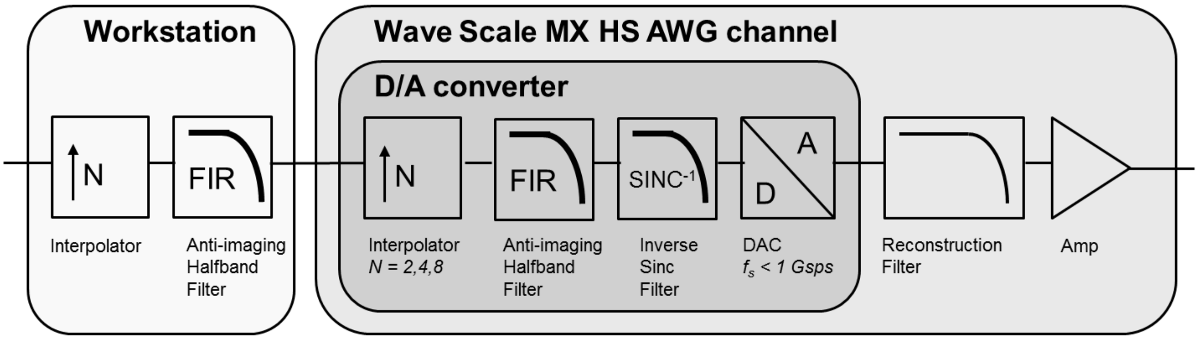


Figure 15: Hard- and software upsampling

Software upsampling resamples the waveform in the workstation CPU, and the upsampled data are stored to the AWG memory. Hardware upsampling uses the three half-band filters of the interpolating D/A converter inside the AWG. Hardware upsampling does not consume additional memory and is limited to the interpolation factor of 8.

Options:

- *off:*
  The D/A converter sample rate is the sampling rate set in the waveform setup (default).

- *memoryOptimized:*
  Only hardware interpolation is enabled, the maximum upsampling factor is 8; no additional waveform memory is consumed

- *imageReduction:*
  The maximum possible upsampling factor is selected to achieve the optimum image reduction. This usually increases waveform memory usage.

- *auto:*
  On WSMX AWGs the system-optimized setup is the same as *imageReduction*.

Example:
```
upsampling = imageReduction;
```

Upsampling can slightly shift the waveform by a few samples, so that the timing relationship of the waveform to other waveforms or digital signals might be affected. If a precise synchronization with other resources is required, upsampling should be turned off.

3.5.5 Reconstruction filters

The output of a D/A converter has to be bandlimited with a low-pass analog filter, called reconstruction filter, to remove images, harmonics and high frequency noise. The WSMX AWGs offer a couple of such filters, and SmarTest uses the mandatory property *bandwidth* to select the best suited one.

Example:

```
bandwidth = 8.9 MHz;
```

WSMX HR AWG

The Wave Scale MX high resolution AWG contains three different analog reconstruction filters with the nominal bandwidths 1, 20 and 80 kHz. They are all 5th order Chebysheff filters with 0.01dB ripple. The settling times of these filters should be considered in the measurement setups:

1 kHz filter: 7 ms
20 kHz filter: 500 us
80 kHz filter: 300 us

1 kHz reconstruction filter

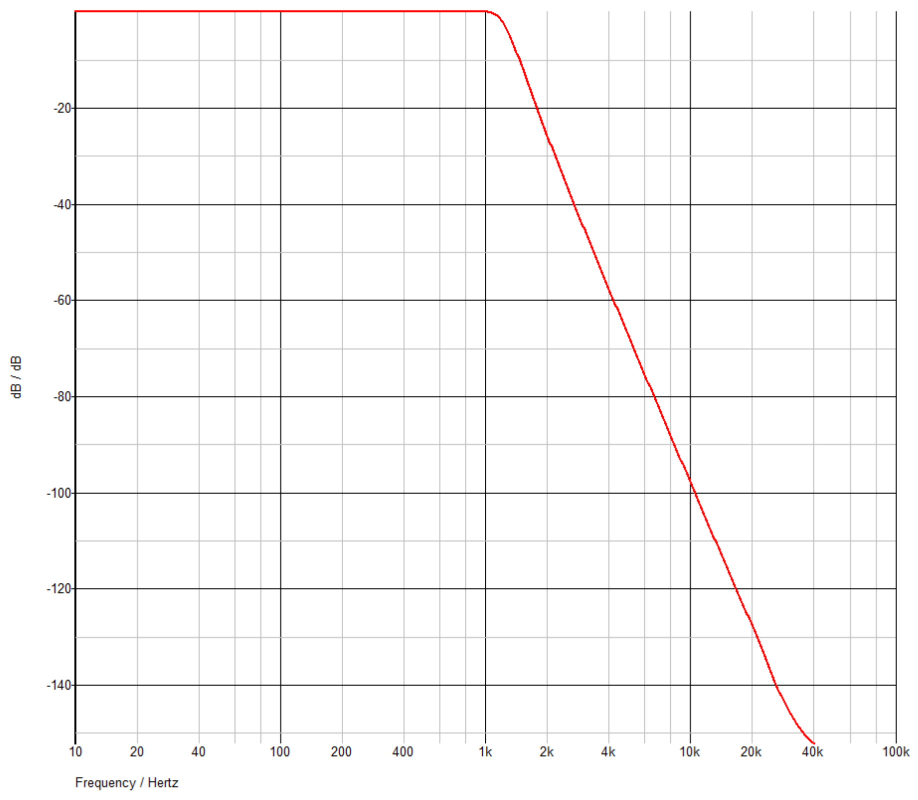


Figure 16: Magnitude response of the 1 kHz reconstruction filter

20 kHz reconstruction filter

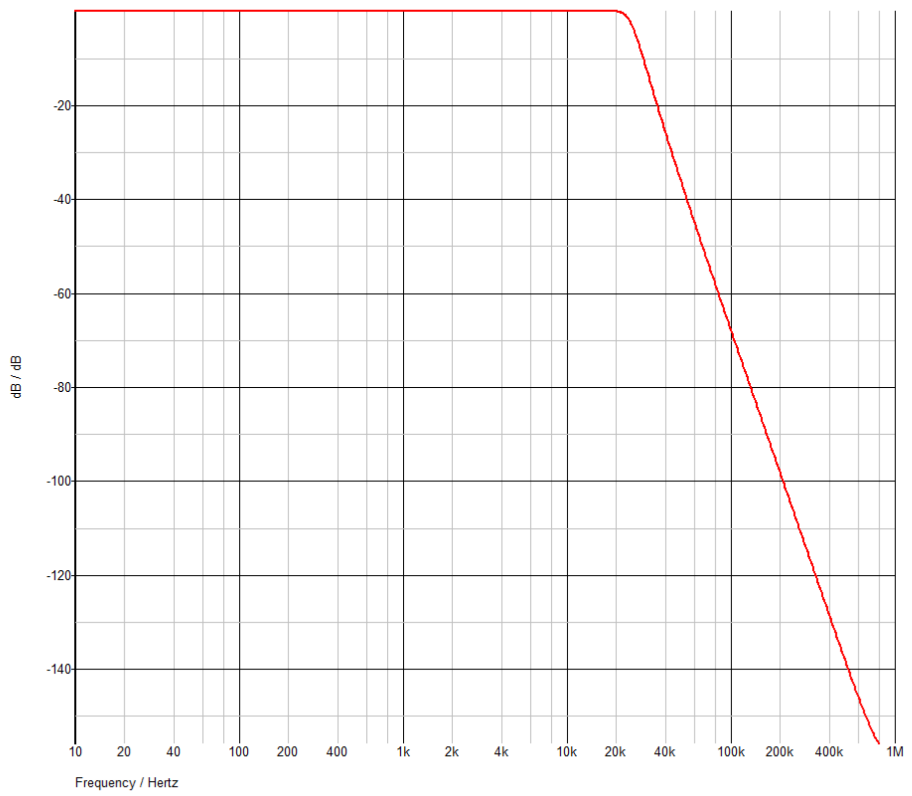


Figure 17: Magnitude response of the 20 kHz reconstruction filter

80 kHz reconstruction filter

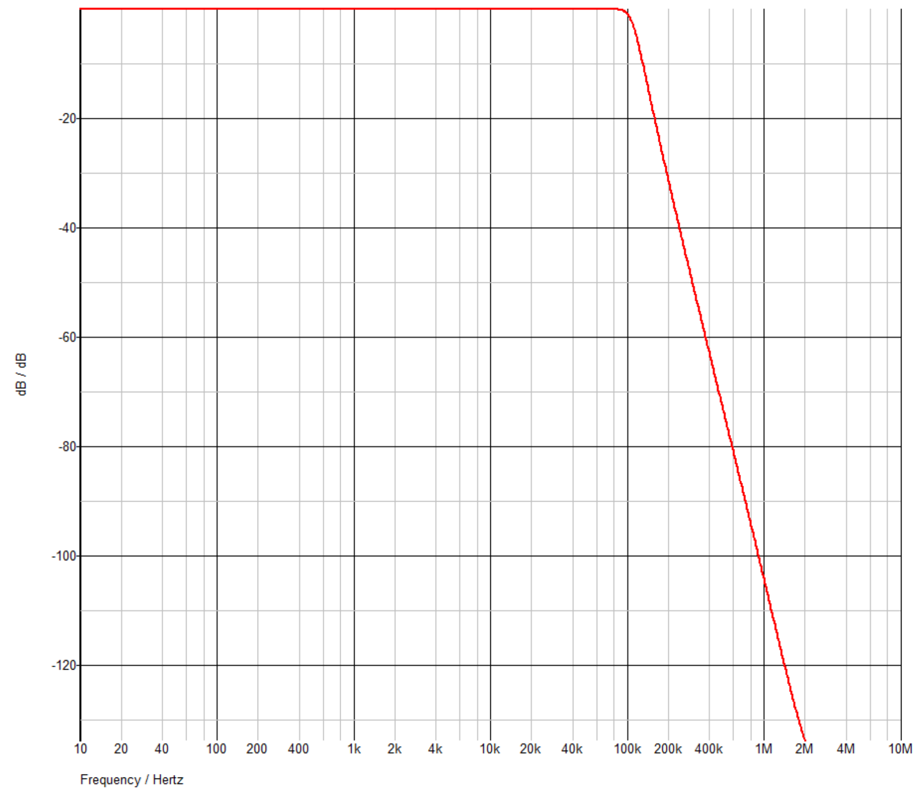


Figure 18: Magnitude response of the 80 kHz reconstruction filter

WSMX HS AWG

The Wave Scale MX high speed AWG has four different reconstruction filters with 5 MHz, 30 MHz, 80 MHz and 200 MHz bandwidth.

5 MHz reconstruction filter

- 5th order Chebyshev filter
- -0.1dB @ 5 MHz, -3dB @ 5.7 MHz
- 0.1 dB passband ripple

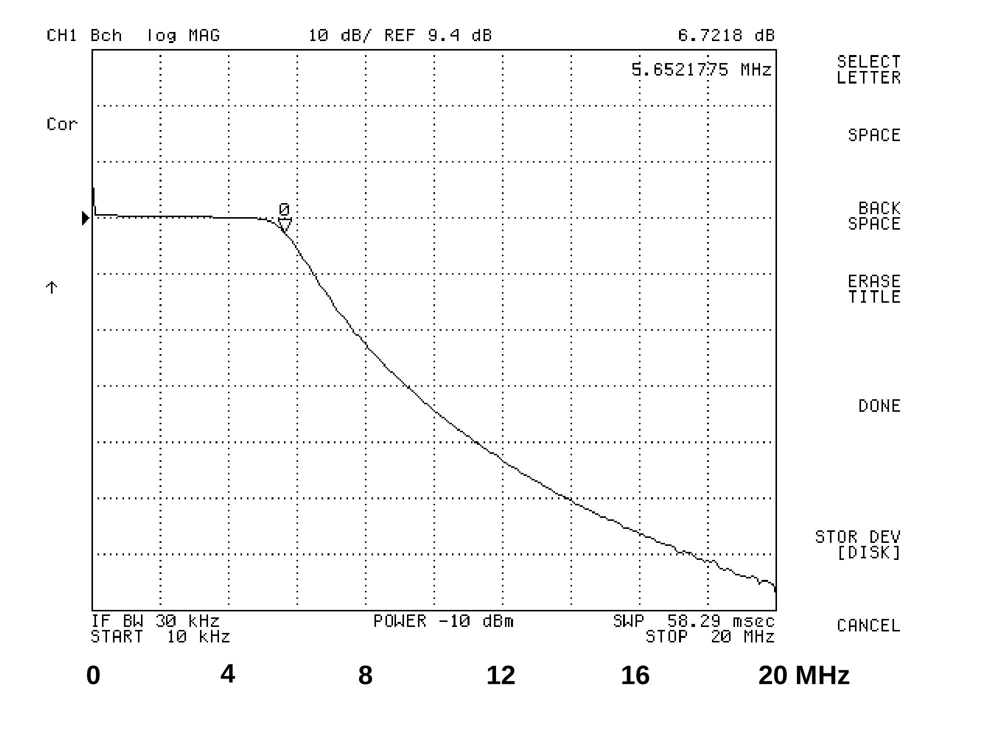


Figure 19: Magnitude response of the 5 MHz WSMX HS AWG reconstruction filter

30 MHz reconstruction filter

- 5th order Chebyshev filter
- -0.1dB @ 30 MHz, -3dB @ 37 MHz
- 0.1 dB passband ripple

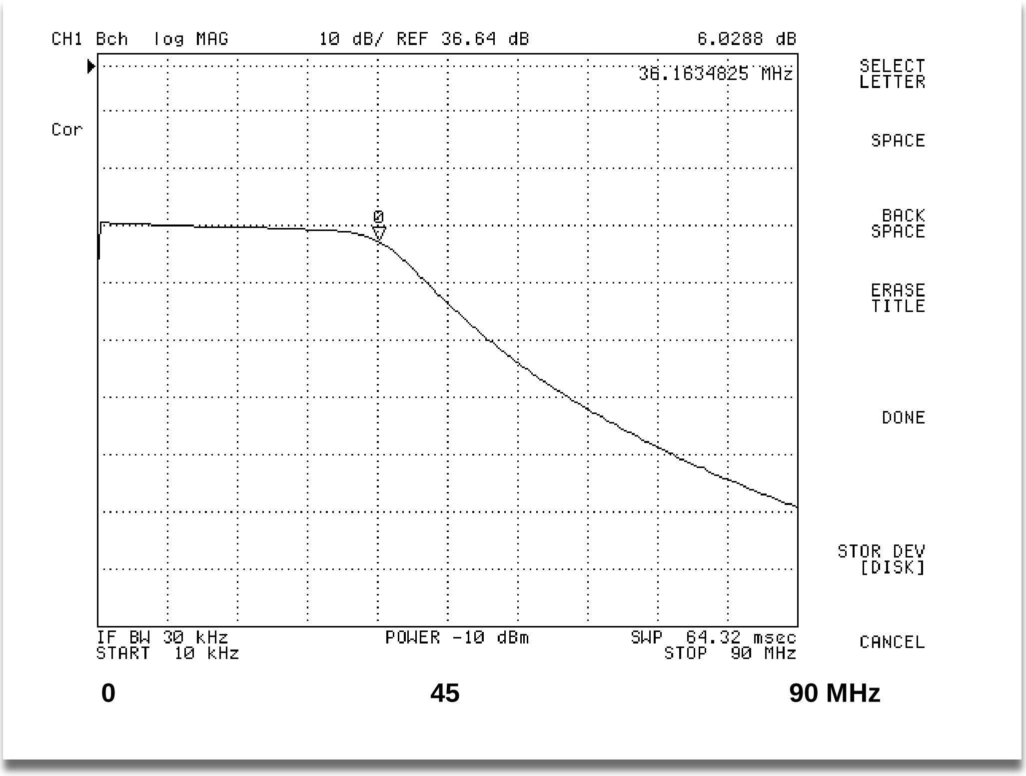


Figure 20: Magnitude response of the 30 MHz WSMX HS AWG reconstruction filter

80 MHz reconstruction filter

- 7th order Cauer filter
- -0.1dB @ 80 MHz, -3dB @ 88 MHz
- Same filter as on MBAV8+ (MCE)

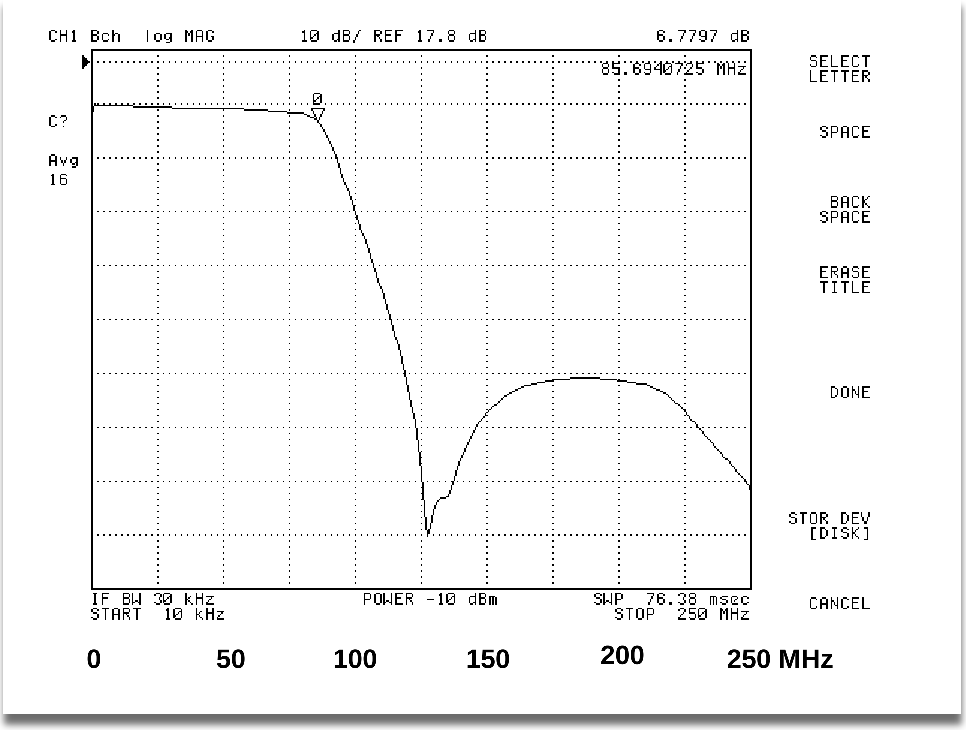


Figure 21: Magnitude response of the 80 MHz WSMX HS AWG reconstruction filter

200 MHz reconstruction filter

- 7th order Cauer filter
- -0.1dB @ 200 MHz, -3dB @ 220 MHz

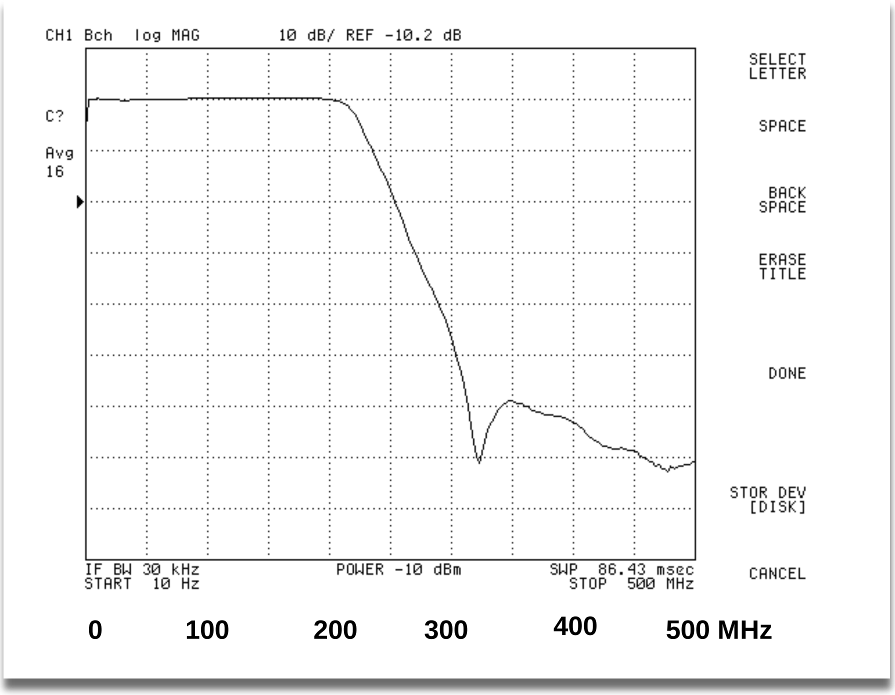


Figure 22: Magnitude response of the 200 MHz WSMX HS AWG reconstruction filter

3.5.6 Termination and Input Impedance

In contrast to power supplies and VI sources, the signal amplitudes and the offsets of AWGs are not regulated. As a consequence the effective amplitude at the DUT depends on the output impedance of the AWG, the wave impedance of the line and the termination on the DUT board or inside the DUT.

The amplitude and DC offset voltage of the signal, which is sourced by the AWG, are compensated such that the effective voltage at the DUT input is the signal amplitude specified by the property *amplitude*. This is only possible if the termination on the DUT board and inside the DUT is specified correctly in the DUT board file.

The amplitude of the AC waveform and the DC offset are compensated independently for the DUT impedance. The AWG output of a WSMX HS unit is always terminated with a 50 Ω series resistor (see also figure 4), that of a WSMX HR unit is always terminated with a 10 Ω series resistor.

A parallel AC termination at the DUT leads to different input impedances for the AC portion and the DC offset of the signal; therefore these values can be set separately.

The default values for the DUT input impedance are as follows:

- No impedance specified:
  AC impedance = DC impedance = high impedance (shown as 1MΩ)
- Only DC impedance specified:
  AC impedance = DC impedance
- Only AC impedance specified:
  DC impedance = high impedance

The user interfaces display high impedance as 1MΩ.

3.5.6.1 No termination

Lines without termination are only suitable for DC and low-frequency signals, and very short trace lengths. Signals in the frequency range of the WSMX HS AWG, traveling from the AWG to the DUT via pogo cables and the DUT board, usually require proper termination. If a transmission line is not properly terminated, the signal is reflected back down the wire.

For non-terminated connections no termination value needs to be specified in the DUT board file; in this case the default values (AC impedance = DC impedance = high impedance) are used.

3.5.6.2 Termination with a pull-down-resistor

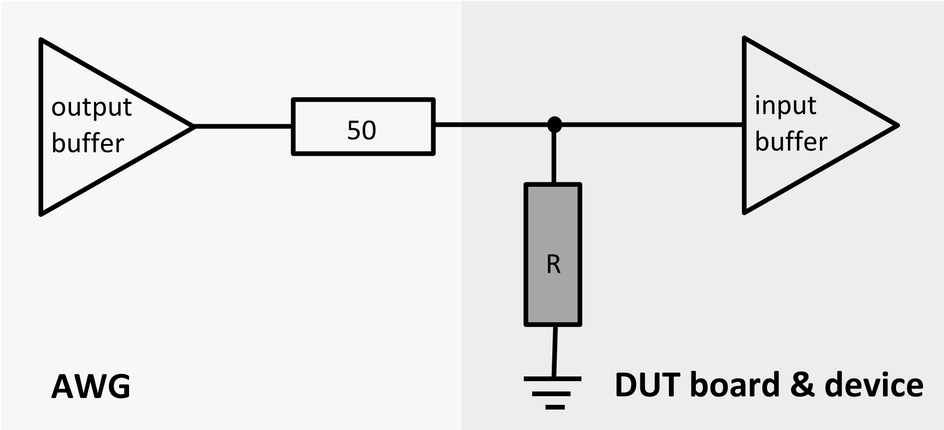


Figure 23: Termination with pull-down resistor

A pull-down resistor can absorb the reflections on the transmission line if the load impedance equals the impedance of the transmission line, usually 50 Ω. However, the power consumption of such a connection is quite high, as the resistor network consumes a lot of DC current, especially when the signal contains a DC offset.

The termination with a pull-down resistor needs to be specified in the DUT board file, so that the amplitude can be compensated correctly. It is sufficient to specify the DC impedance *dutImpedanceDC;* the default AC impedance equals the DC impedance.

Example:

```
dutboard dutboard4site {

sites = 4;

     signal RX {
          site 1  { pogo = 10116; dutImpedanceDC = 50Ohm;}
          site 2  { pogo = 10114; dutImpedanceDC = 50Ohm;}
        …
    }
}
```

3.5.6.3 Parallel AC termination

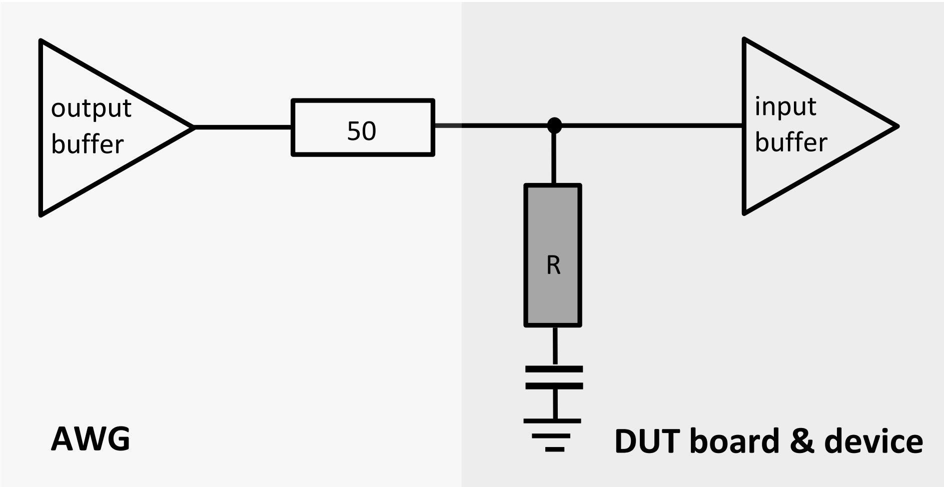


Figure 24: Parallel AC Termination

The best termination for most applications is the parallel AC termination. The additional series capacitor blocks the DC average current and thus reduces the DC power consumption to a minimum. In addition the AWG yields the best THD performance with this kind of termination.

The DUT's input impedance of the parallel AC termination is different for the AC and the DC portion of the AWG output voltage. The cutoff frequency is a function of the resistor and the capacitor:

$$f_c = \frac{1}{2\,\pi\,\tau} = \frac{1}{2\,\pi\,RC}$$

The cutoff frequency should not be higher than one tenth of the lowest used frequency component of the transmitted signal. Assuming the resistor R matches the transmission line impedance, which is usually 50 Ω, the capacitor should be larger than

$$C \geq \frac{1}{100\,\pi\,f_a}$$

It is sufficient to set the AC impedance; the default DC impedance is high impedance in this case.

Example:

```
dutboard dutboard4site {

sites = 4;

    signal RX {
        site 1  { pogo = 10116; dutImpedanceAC = 50Ohm;}
        site 2  { pogo = 10114; dutImpedanceAC = 50Ohm;}
     …
   }
}
```

3.5.6.4 Termination of differential connections

For differential connections, the DUT input impedance is the equivalent single-ended impedance into ground: If the input impedance of a certain signal is set to 50 Ω, 100 Ω input impedance is assumed in differential mode.

The DC input impedance of a differential connection without center ground is always high-Z, even without a series capacitor.

3.5.7 Amplitude

Use the property *amplitude* to set the signal amplitude in Volts peak (Vp).
Default: 1 Vp

Example:
```
Amplitude = 1.2 V;
```

The amplitude of an AC waveform is set by the property *amplitude*. All amplitudes of signals, driven by a Wave Scale MX AWG or captured by a Wave Scale MX digitizer, are defined as peak amplitudes, and the unit is Volts peak (V_{peak}). This corresponds to the length of the complex phasor and is commonly used in telecommunications, audio system measurements and other areas, where signals swing above and below an offset voltage or ground.

The amplitude always refers to a single channel versus offset, even in differential mode, so that the resulting differential amplitude is two times the defined amplitude.

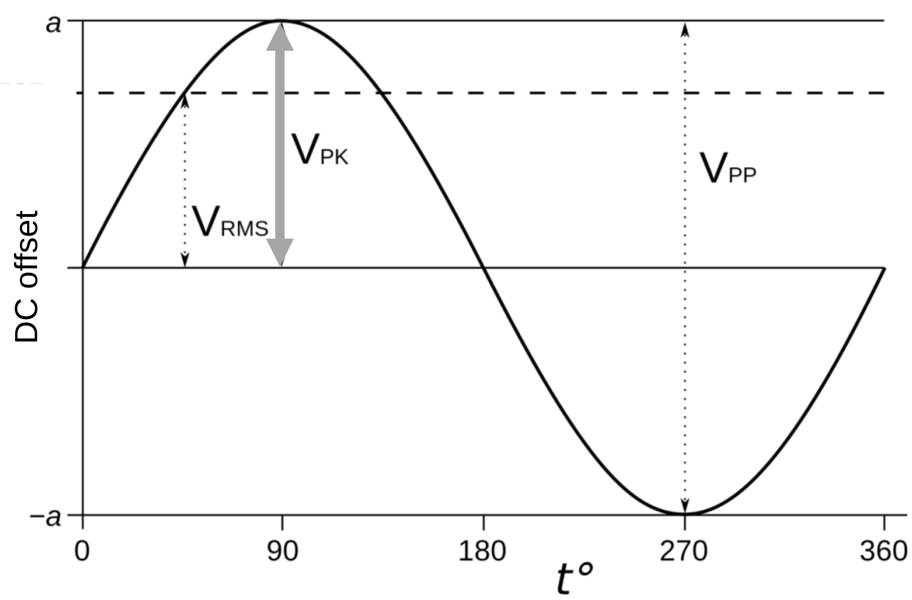


Figure 25: The amplitude of an AWG signal

The appropriate range is selected by the system software such that the D/A converter operates with the maximum resolution. In most cases the user does not need to care about the range. Only in cases, where switching ranges leads to undesired effects, the user can set a fix range with the *amplitudeRange* property. All setups with the same *amplitudeRange* setting operate with identical output amplifier settings.

WSMX HR
The WSMX HR AWG has eight different ranges:
0.75 V_p, 1.5 V_p, 2.6 V_p, 3.35 V_p, 5.2 V_p, 6.7 V_p and 10.4 V_p

The amplitude of the WSMX HR AWG might drift slowly after changing the sampling rate.

3.5.8 Offset

The properties *offset* and *offsetNeg* describe the DC content of the signal separately for each leg. The offset voltage of an AWG channel is set by the property *offset*. For single-ended connections it sets the DC offset of the positive leg, for differential connections it sets the DC offset of both the positive and the negative leg. If the negative leg should get a different offset value, use the property *offsetNeg*.

Offset voltages are compensated for the input termination of the DUT.

DC offset in a differential setup significantly increase the harmonics.

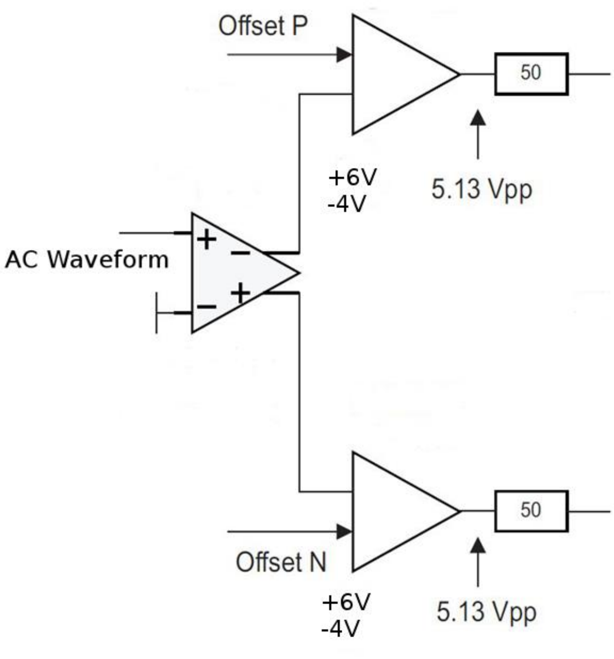


Figure 26: Schematics of the AWG output

Use the property *offset* to set the DC offset in Volts (V). Use the property *offsetNeg* to set the DC offset of the negative leg, in case different offsets are required for the positive and the negative leg.

Default value of *offset*: 0 V
Default value of *offsetNeg*: *offset*

3.5.9 Target Frequency

Use the property *targetFrequency* to increase the accuracy of the signal amplitude. Setting the target frequency is mandatory in IQ mode (instrument option *IQ* is set). This property has no effect on high resolution modules.

The default value is waveform dependent:

- For DC waveforms and ramps: 1 kHz
- For sine and square waves: signal frequency
- For chirps: (start frequency + stop frequency) / 2
- For multi-tones: (lowest frequency + highest frequency) / 2
- For noise and sample-based waveforms: bandwidth / 2.

3.5.10 Optimization

The output signals of the HR AWG can be optimized for either low noise or low harmonic distortion. Use this option to optimize the signal according to the requirements of your measurement and the figures of merit you are going to test. The available values are *distortion* and *noiseFloor*.
This option is only available on the HR AWG.
Default: *noiseFloor*

Example:
```
optimization = distortion;
```

3.6 AWG Setup with the Device Setup API

In addition to the SmarTest Setup Format (SSF), SmarTest's own domain specific language, AWGs and digitizers can also be set up using the Device Setup API, the Java test method interface to the instruments. The Device Setup API provides a complete set of functions to set up measurements. The example below shows a complete AWG setup.

```
package Setup_Examples;

import xoc.dsa.DeviceSetupFactory;
import xoc.dsa.IDeviceSetup;
import xoc.dsa.ISetupAwg;
import xoc.dsa.ISetupAwg.ISourceWaveform;
import xoc.dsa.ISetupWaveformSine;
import xoc.dta.TestMethod;
import xoc.dta.measurement.IMeasurement;

public class LineInput extends TestMethod {

    public double SignalFrequency = 997;
    public double AWGSampleRate   = 1000E3;
    public double AWGAmplitude    = 0.5;
    public long   AWGSamples      = 10000;

    public IMeasurement measurement;

    @Override
    public void setup(){

        IDeviceSetup ds = DeviceSetupFactory.createInstance();

        ///////////////////////////////////////////////////////////////
        // Generate the AWG waveform
        ///////////////////////////////////////////////////////////////

        ISetupWaveformSine waveform = ds.addWaveformSine("Sine_Waveform");

        waveform.setFrequency  (SignalFrequency);
        waveform.setSampleRate (AWGSampleRate);
        waveform.setSamples    (AWGSamples);

        ///////////////////////////////////////////////////////////////
        // Setup the AWG instrument
        ///////////////////////////////////////////////////////////////

        ISetupAwg awgSetup = ds.addAwg("LINE_MIC1_P+LINE_MIC2_P");
        awgSetup.setConfigOptionDifferential();

        ISourceWaveform Source_Waveform =
                        awgSetup.sourceWaveform("Source_Waveform");
        Source_Waveform.setWaveform("Sine_Waveform");
        Source_Waveform.setAmplitude(AWGAmplitude);
        Source_Waveform.setOffset(1.0);
        Source_Waveform.setBandwidth(SignalFrequency);
        Source_Waveform.setRepeatCount(10);
```

```
        ///////////////////////////////////////////////////////////////
        // Generate the operating sequence:
        ///////////////////////////////////////////////////////////////

        ds.parallelBegin("Grp1");
        {
            ds.sequentialBegin("AWG");
            {ds.actionCall(Source_Waveform);}
            ds.sequentialEnd();

        }
        ds.parallelEnd();

        measurement.setSetups(ds);

    }

    @Override
    public void execute() {

        measurement.execute();
        // (…)

    }
}
```

4. Digitizer Setup

The SmarTest8 digitizer instrument model is used to set up the WSMX digitizers.

4.1 DUT Board

All signals are assigned to pogo numbers in the DUT board file, together with the DUT board properties.

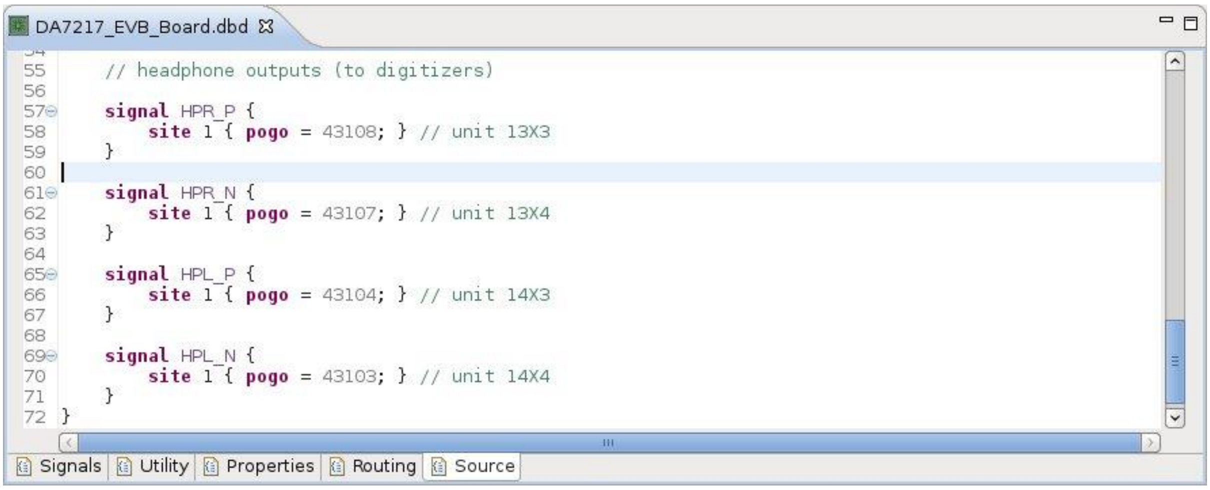


Figure 27: Digitizer pins in the DUT board file

4.2 Digitizer instrument properties

These properties apply to all actions within one specification and cannot be changed during a measurement execution. All properties are optional.

4.2.1 connect

This property specifies if the digitizer instrument should connect before the measurement execution. If set to *false*, the instrument is not connected during the measurement.
Default: *true*

Example:

```
connect = true;
```

4.2.2 disconnect

This property specifies whether to disconnect the digitizer instrument after the measurement execution. If set to false, the instrument remains connected after the measurement.
Default: *false*

Example:

```
disconnect = true;
```

4.2.3 delay

To fine-tune the synchronization between the digitizer and the other instruments, or to discard initial samples, which might contain transients, the actual start of an action can be delayed with the delay property.
Default: 0 ns

Example:
```
delay = 9.1ns;
```

4.2.4 superPeriod

This property sets a super period to align all signals in this instrument's phase domain.
Default: Least common multiple (LCM) of all periods.

Example:
```
superPeriod = 103.1ns;
```

4.2.5 alarm

The digitizer supports three alarms: the power budget alarm is issued, when the instrument draws too much current (WSMX-HR only). The SPU alarm indicates a fatal condition of the signal processing unit (SPU). These two alarms are fatal and not configurable. The over-range alarm, which indicates that the captured waveform exceeded the digitizer range, is configurable and can be turned on or off.

Example:
```
alarm.overrange = enabled;
```

4.2.6 compensation

Use this property to set the quality of the amplitude compensation. Currently only standard compensation is supported.

Example:
```
compensation = standard;
```

4.3 Digitizer instrument options

Like the instrument properties the instrument options apply to all actions of a pin or pin group within one specification and cannot be changed during a measurement run. In the specification file they are located inside the *setup* block together with the instrument properties.

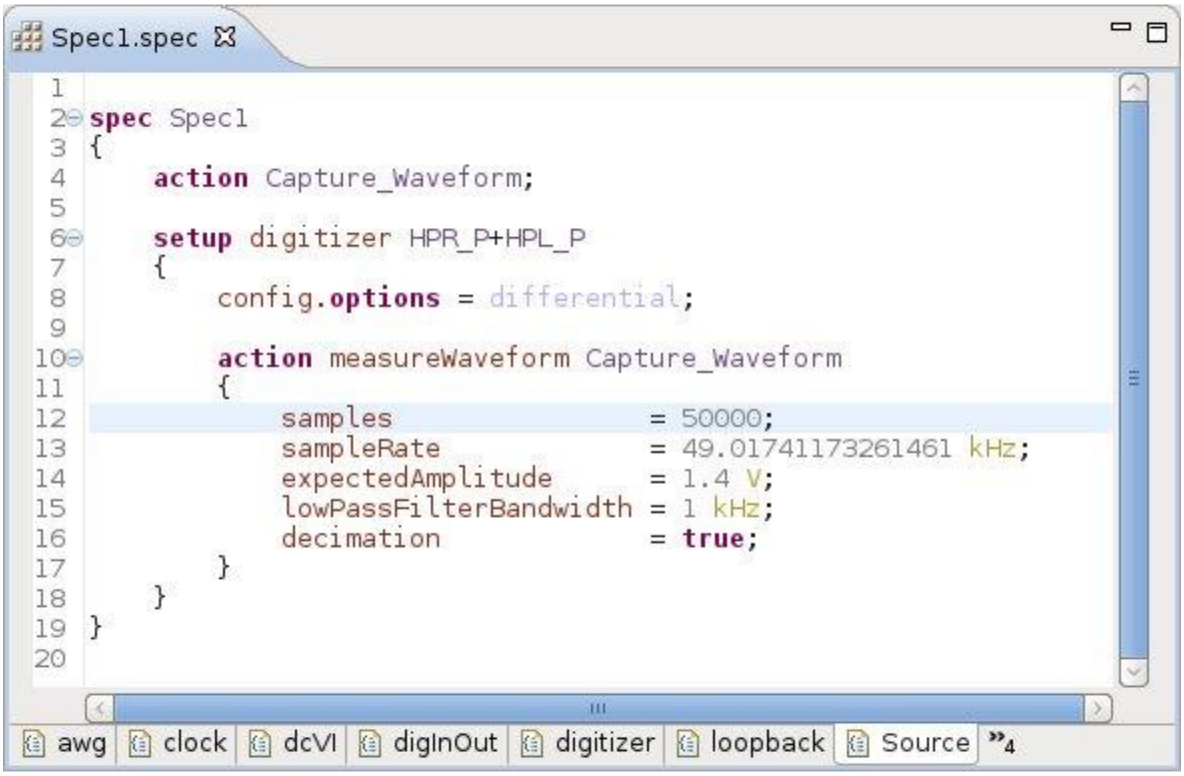


Figure 28: SmarTest8 specification file

4.3.1 Differential connection

The setup of differential connections on Wave Scale MX channels is very similar to single-ended connections. The negative leg of the differential pin pair does not require a separate setup. However, it needs to be declared in the DUT board file.

The property *differential* enables differential connectivity. The negative leg of the differential pair is determined by the hardware. An error is thrown if this signal is not defined in the DUT board file. By default, all connections are single-ended.

Example:

```
config.options = differential;
```

On the WSMX HR digitizer the offset voltage of the negative leg can differ from that of the positive leg and is set by the property *offsetNeg*. The default value is the offset voltage of the corresponding positive leg *offset*.

Note that the property *expectedAmplitude* always refers to a single channel versus offset, so that in differential mode the resulting differential amplitude is two times this amplitude.

4.3.2 Internal Cross-Connect

Use the property *internalCrossConnect* to internally connect the positive AWG output to the positive digitizer input of the same unit without a load board connection (internal loop back). Internal loopback is always single-ended. To establish the connection the option must be set for both the AWG and the digitizer.

Example:
```
config.options = internalCrossConnect;
```

4.3.3 Ground sense

In single-ended mode the accuracy of the forced waveform can be improved by sensing the ground plane with the negative leg of the differential pair. This requires a connection between the negative leg and the ground plane on the DUT board. It is recommended to use this option in single-ended setups.

On the WSMX HR digitizer the option *groundSense* is basically a differential setup with the negative leg connected to ground, and 0V DC offset (*offset Neg* = 0 V). This highly recommended option is only available in single-ended setups on WSMX HR digitizer modules.

Use the property *gndSense* to activate ground sensing.

Example:
```
config.options = gndSense;
```

4.3.4 Resource Sharing

One digitizer unit can test up to four different sites in single-ended mode, or two sites in differential mode, without external relays. Just assign the four pogo connectors X1, X2, X3 and X4 (single-ended mode) or the two differential pairs X1/X2 and X3/X4 (differential mode) to different sites in the dutboard file. The measurements are executed sequentially or interlaced. The phase of the captured signal on the negative leg is inverted.

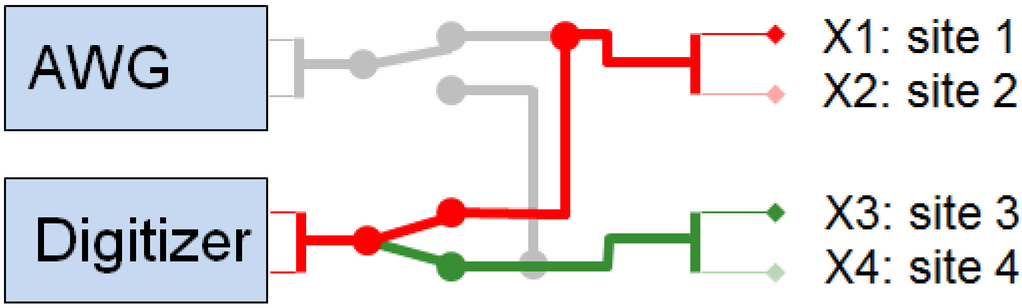


Figure 29: Resource sharing on the WSMX digitizer

In sequential execution mode, which is the default, one complete measurement run is executed per site. In interlaced execution mode only one measurement run is executed, and the digitizer switches between the sites just for the actual digitizer measurement. Make sure that the devices under test drive the signal, which is going to be measured, for the extended time.

Example:

```
config.options = siteInterlacingOn;
```

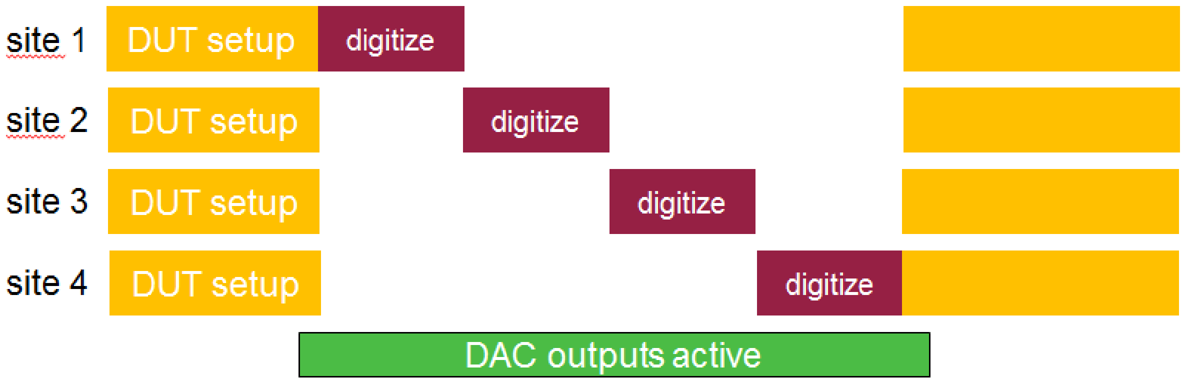


Figure 30: Interlaced execution of a digitizer action

4.4 Properties of the *measureWaveform* action

Action properties are located inside the *action* block of the specification file. They are only effective for one specific action.

4.4.1 Input Impedance and center ground

Choose the input termination of the digitizer with the property *inputImpedance*. Connect the center ground in differential mode with the property *centerGnd*.

WSMX HR digitizer

These properties are not applicable for the WSMX HR digitizer. The inputs of this digitizer have high impedance (high-Z) since they are not connected to ground via a termination resistor. For specific terminations a resistor can be placed on the DUT board.

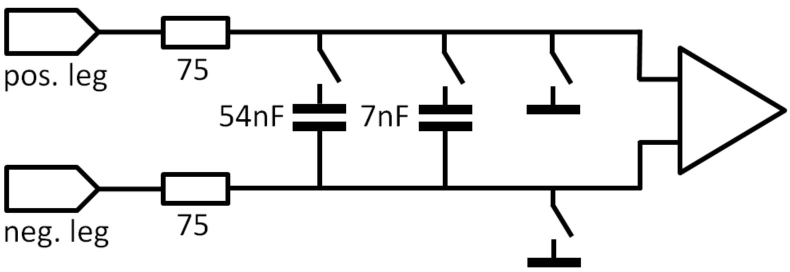


Figure 31: Schematics of the WSMX HR digitizer input

WSMX HS digitizer

In single-ended mode the WSMX HS digitizer input can be terminated with either 50 Ω or 100 kΩ input impedance versus ground. In differential mode the center tap of the 100 Ohms termination can be optionally connected to ground. In this case the digitizer's input impedance is different for the AC portion and the DC offset of the DUT's output signal.

Example:

```
inputImpedance = 100 kOhm;
centerGnd      = true;
```

For differential connections the DUT input impedance is the equivalent single ended impedance into ground: to set 100 Ω input impedance in differential mode, set *inputImpedance* to 50 Ω.

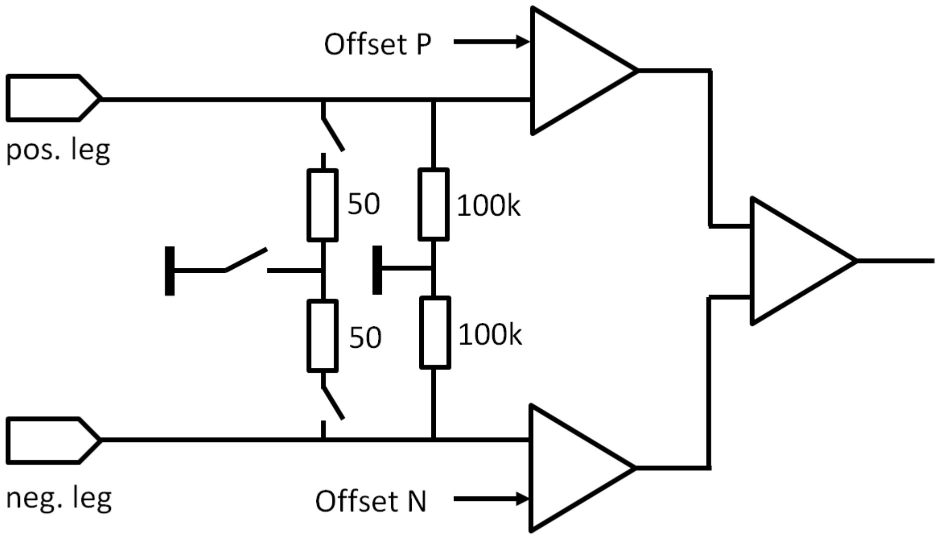


Figure 32: Schematics of the WSMX HS digitizer input

4.4.2 Voltage Range

The voltage range of the digitizer is determined by the system according to the digitizer property *expectedAmplitude;* the unit is Volts peak (V_p). The property *expectedAmplitude* always refers to a single channel versus ground, even in differential mode, so that it has to be set to half the value of the differential amplitude.

Set the property *expectedAmplitude,* which sets the range, to a value close to the maximum expected amplitude, without much of a margin. The automatic range selection already adds a margin to the *expectedAmplitude.*

Example:

```
expectedAmplitude = 625 mV;
```

WSMX HR digitizer

Input Voltage Ranges: 260 mVp, 520 mVp, 1.3 Vp, 2.6 Vp, 5.2 Vp, 10.4 Vp

The voltages at the digitizer inputs are offset-dependent.

differential:

$-V_{range} + V_{off} \le V_{diff} \le +V_{range} + V_{off}$

$-5.2\ V + V_{off} \le V_P \le +5.2\ V + V_{off}$

$-5.2\ V - V_{off} \le V_N \le +5.2\ V - V_{off}$

single-ended P:

$-V_{range} + V_{off} \le V_P \le +V_{range} + V_{off}$

single-ended N:

$-V_{range} + V_{off} \le -V_N \le +V_{range} + V_{off}$

WSMX HS digitizer

The programmable gain amplifier of the digitizer input allows for dynamic ranging with voltage ranges from 62.5 mV_{peak} to 2 V_{peak}. The system software selects the appropriate input range based on the setup property *expectedAmplitude*. It is defined as peak amplitude, the unit is Volts peak (V_{peak}) and refers to a single channel versus ground, even in differential mode (see also figure 3).

The dynamic performance of the digitizer is range-dependent, and also the harmonic distortion, but to a smaller degree.

4.4.3 DC Offset

Use the property *offset* to set the DC offset voltage of the positive and the negative leg of the digitizer input to the same value. Use the property *offsetNeg* to set the offset voltage of the negative leg to a different value.

Example:
```
offset    = 1.1 V;
offsetNeg = 0.9 V;
```

In a differential connection the offset D/A converter of the digitizer is set to

V_{off} = *offset – offsetNeg*

and the common mode is

V_{cm} = 0.5 * (*offset* + *offsetNeg)*

WSMX HR
Common mode offset is not supported in the 10.4 Vp range (*expectedAmplitude* > 2.6Vp)

4.4.4 Anti-aliasing filter

Use the property *lowPassFilterBandwidth* to set the bandwith of your signal. This property is mandatory. The SmarTest software chooses the appropriate anti-aliasing low pass filter for that bandwidth. This property has no effect on WSMX HR units.

Example:

```
lowPassFilterBandwidth = 26 MHz;
```

WSMX HR digitizer

The WSMX HR digitizer has a fix 350 kHz (-3dB point) anti-aliasing filter in its input signal path. For reference: MBAV8+ (MCE) has a 50 kHz filter and a digital low pass filter which scales with the sample rate.

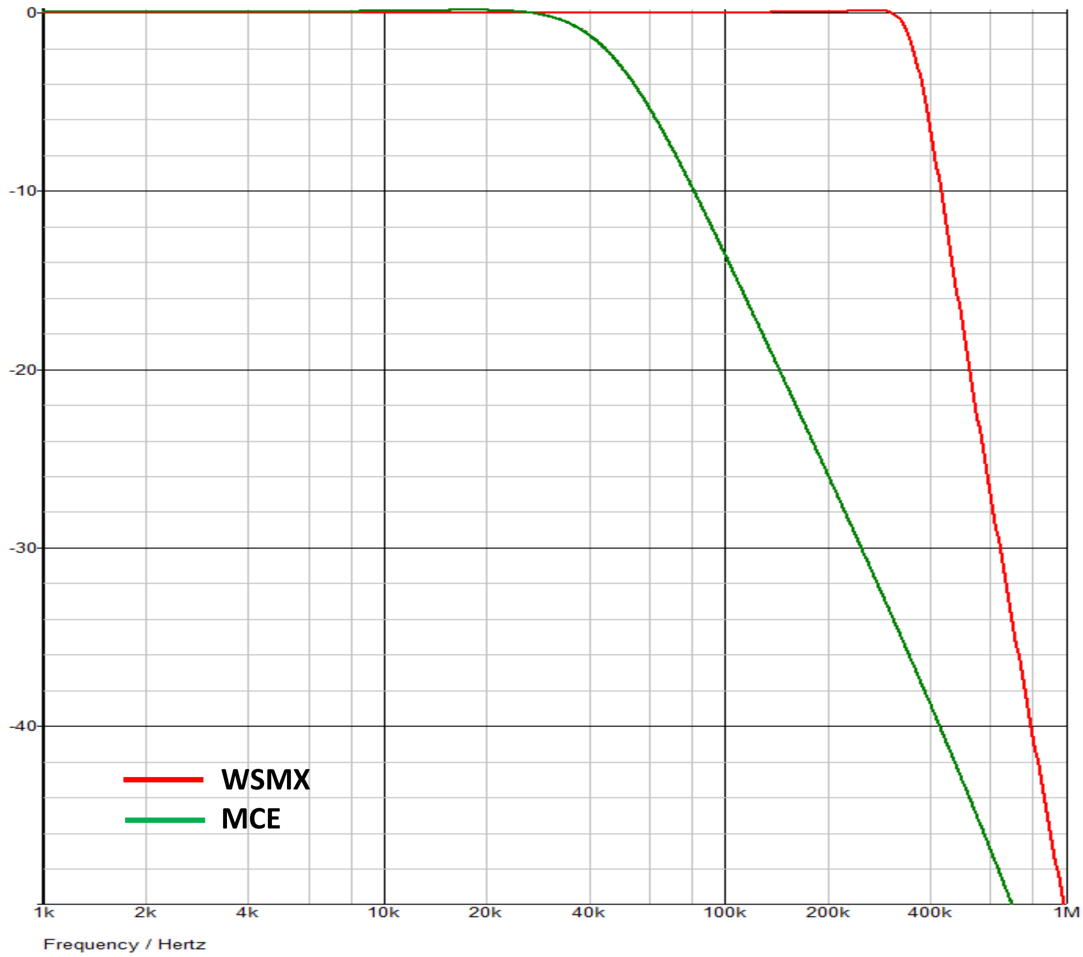


Figure 33: Magnitude responses of the 350 kHz WSMX HR digitizer filter (red) and the 50 kHz MCE LF digitizer filter (green)

This filter is always active; the digitizer property *lowPassFilterBandwidth* has no effect.

WSMX HS digitizer

The WSMX HS digitizer has a 100 MHz anti-aliasing filter, which is bypassed if the property *lowPassFilterBandwidth* is set to value higher than 100 MHz.

4.4.5 Samples

Use the property *samples* to set the number of samples to be returned. This property is mandatory. The property *samples* always refers to the number of samples which is returned by the result accessor. The A/D converter of the digitizer might capture a lot more samples in case decimation or averaging is turned on.

Example:
```
samples = 1024;
```

4.4.6 Sample Rate

Use the property *sampleRate* to set the sample rate of the waveforms to be returned. This property is mandatory. *sampleRate* always refers to the samples which are returned by the result accessor; the A/D converter of the digitizer might capture at a higher rate in case decimation or averaging is turned on.

Example:
```
sampleRate = 48 kHz;
```

WSMX HR

WSMX HR supports sample rates from 1 sps to 1.5 Msps. The analog bandwidth of the WSMX HR digitizer is 350 kHz, taking into account all analog and digital filters. To prevent aliasing, it is highly recommended to use sampling rates between 700 ksps and 1.5 Msps to prevent signal content above the Nyquist frequency from being aliased into the 1st Nyquist band without being filtered.

WSMX HS

WSMX HS supports sample rates from 250 ksps to 250 Msps. To prevent aliasing, and to minimize noise, the sample rate should be in the range between 200 and 250 Msps, so that the 100 MHz anti-aliasing filter becomes effective.

Lower sampling rates, however, reduce the distortion of the A/D converter and yield better THD. For example, a 5 MHz tone with 100 Msps yields a THD of 87 dBc, while at 250 Msps it might yield only 77-80 dBc.

4.4.7 inputCapacitance

Delta-Sigma D/A converters using noise shaping techniques can create a high amount of high frequency noise. If unfiltered, it can cause slew-rate overdrive of the instrument’s input amplifiers, even though the signal level is within the specification. As the anti-aliasing filter is located behind the input amplifiers, it does not filter out this high frequency noise.

A capacitor at the digitizer input can limit the slew rate and remove the high frequency noise. The capacitor forms a low pass filter together with the 75Ω input impedance and the impedances in the remaining signal path, e. g. on the DUT board. This simple RC filter rolls off the frequency response at 6 dB per octave above the cutoff frequency, which is

$$f_c = \frac{1}{2\pi \mathrm{RC}}$$

This feature is only available on WSMX HR digitizers.

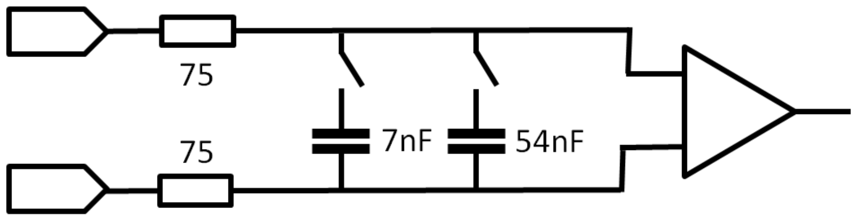


Figure 34: Capacitors at the HR digitizer input

Use the property *inputCapaciance* to connect a capacitor at the digitizer input. Available capacitors are *none, C7nF* and *C54nF*. This property is optional. The default value is *none* (no capacitor connected).

Example:
```
inputCapacitance = C7nF;
```

4.4.8 Decimation

Decimation is the process of reducing the sampling rate of a signal. Decimation utilizes filtering to mitigate aliasing distortion, which can occur when simply downsampling a signal. Prior to decimation the signal is oversampled. This technique is commonly used to reduce digitizer noise and aliases in the captured signal. Decimation only reduces the noise which is produced by the digitizer, and not the noise of the incoming signal itself; therefore it is legitimate to use it also for noise measurements.

By default the decimation is turned off on WSMX digitizers (*decimation = false*), and the A/D-converter of the digitizer instrument samples at the user specified sample rate. Use the property *decimation* to turn it on (*decimation = true*). It is not necessary to modify the sampling setup; SmarTest itself computes the optimum oversampling and decimation setup.

When decimation is turned on, the usable bandwidth of the obtained spectrum is reduced. The upper part of the spectrum is not compensated and may contain aliases. Within the usable bandwidth, all amplitudes are precise and aliases are suppressed.

The *spectrum()* function of the *MultiSiteWave* class returns the entire spectrum, including the unusable part. However, the values returned by the member functions of the spectrum object (e. g. *snr()*, *snd()* or *thd()* for SNR, SND and THD computation) are based on the usable bandwidth; frequencies outside the usable bandwidth are not taken into account for the calculations.

Example:
```
decimation = true;
```

Decimation reduces aliasing
At higher sampling rates all aliases are shifted to higher frequencies. The anti-aliasing low-pass filters of the digitizers suppress these aliases to a higher degree.

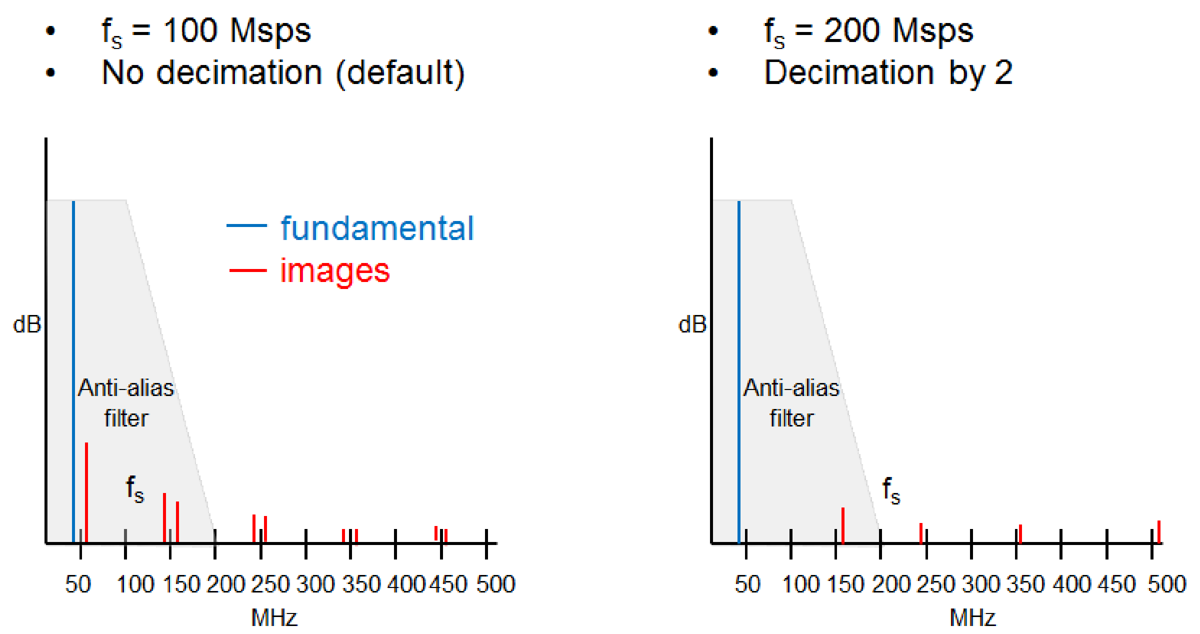


Figure 35: Alias reduction through decimation

Noise reduction through decimation

Decimation reduces the noise level by approximately *$3*log_2$(oversampling ratio) dB*.

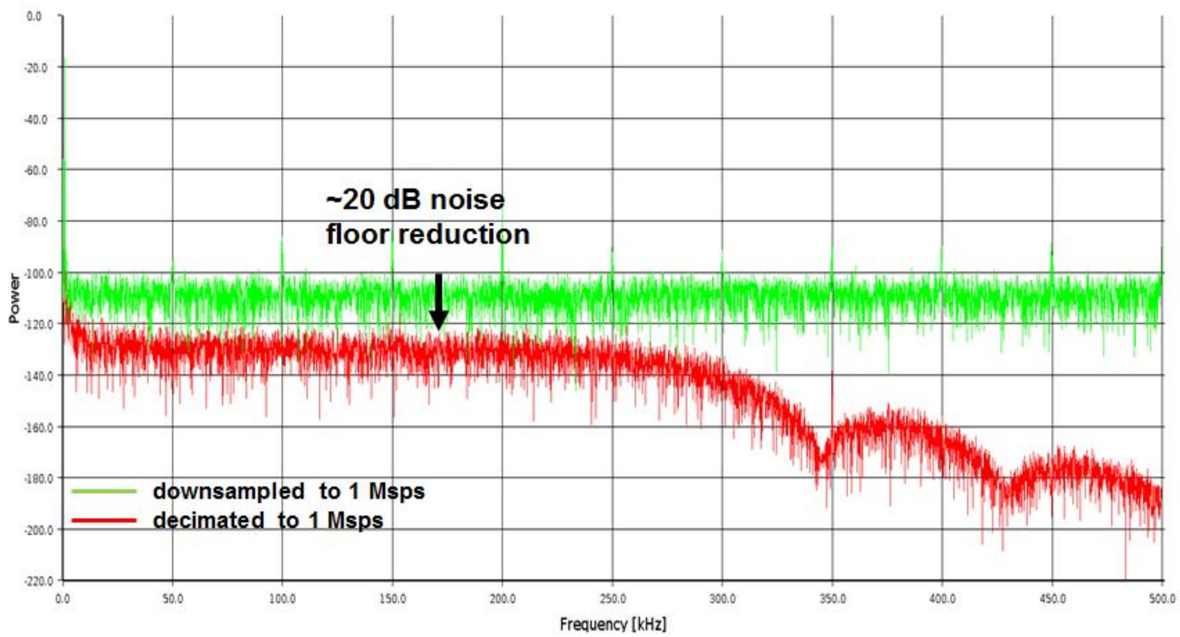


Figure 36: 1 kHz-waveform decimated by a factor of 250 to 1 Msps, thereby reducing the noise floor by more than 20 dB.

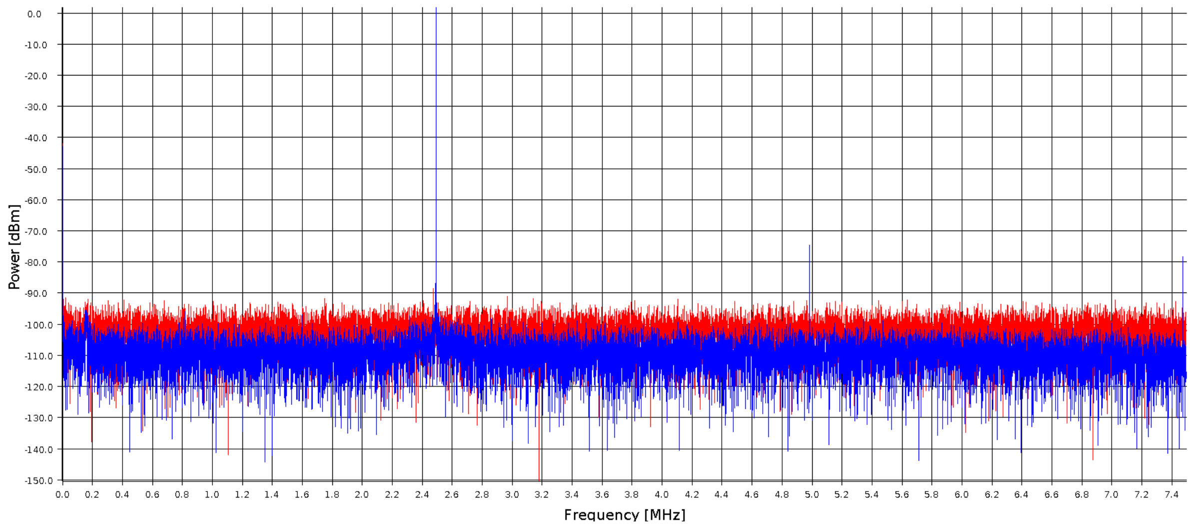


Figure 37: SNR of a 2.5 MHz signal reduced by 8 dB from 58.0 dBc without to 66.2 dBc with decimation

The low pass filter of the decimator reduces the usable bandwidth of the spectrum. Within the usable bandwidth, the frequency response is flat, and aliases are significantly suppressed. Beyond this bandwidth the spectrum contains aliases, and the magnitudes may include errors.

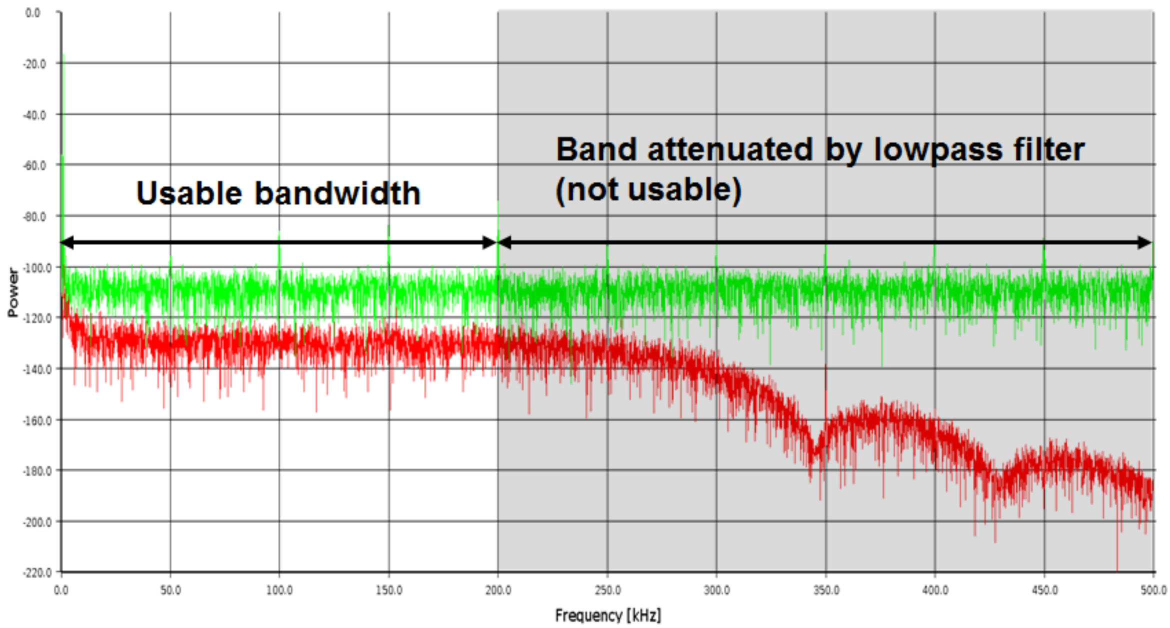


Figure 38: Usable bandwidth of a decimated waveform

WSMX HR

The analog bandwidth of the WSMX HR digitizer is 350 kHz. To prevent aliasing, the sampling rate of the A/D-converter should be in the range between 700 and 1500 ksps. If the selected sampling rate is lower, decimation should be turned on as otherwise the signal content above the Nyquist frequency ($f_s/2$) is aliased into the first Nyquist band without being filtered.

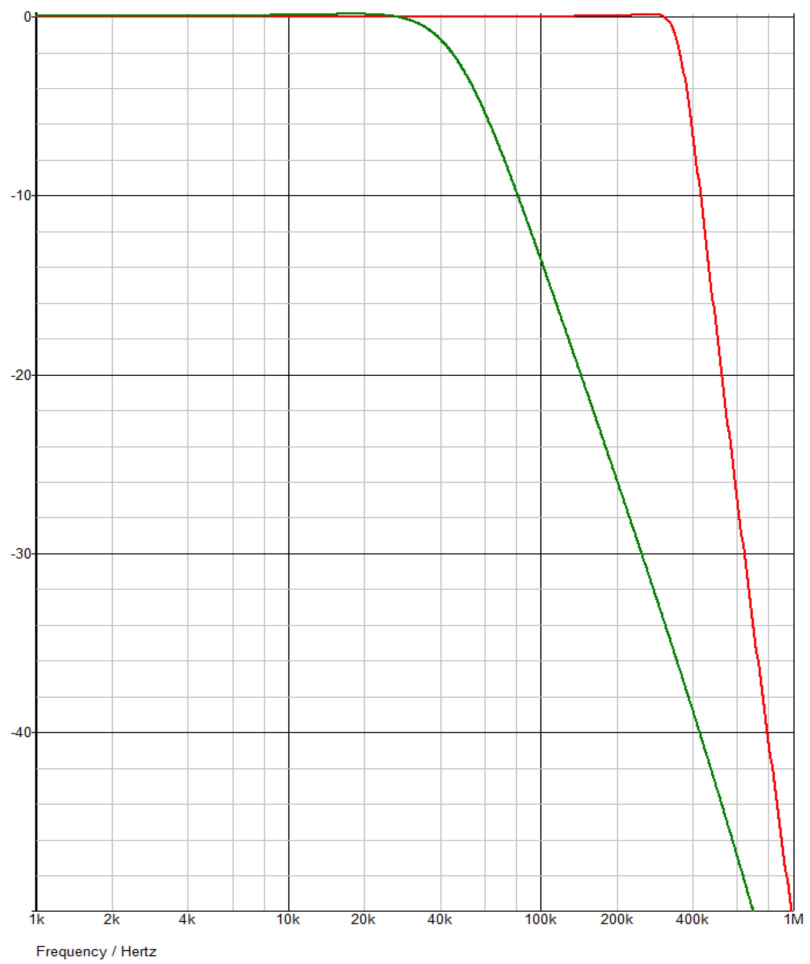


Figure 39: Frequency response of the anti-aliasing filter of WSMX HR (red) and MBAV8+ (green)

The usable bandwidth of a decimated waveform captured with the WSMX HR digitizer is 23/50 or 46% of the sample rate, which corresponds to 92% of the Nyquist frequency. The minimum sample rate of the digitizer is 1 sps without decimation and 6.25 ksps with decimation.

WSMX HS

The analog bandwidth of the anti-aliasing filter of the WSMX HS digitizer is 100 MHz. For optimum noise and alias reduction, the sample rate of the A/D converter should be in the range between 200 and 250 Msps. If the selected sampling rate is lower, decimation should be turned on.

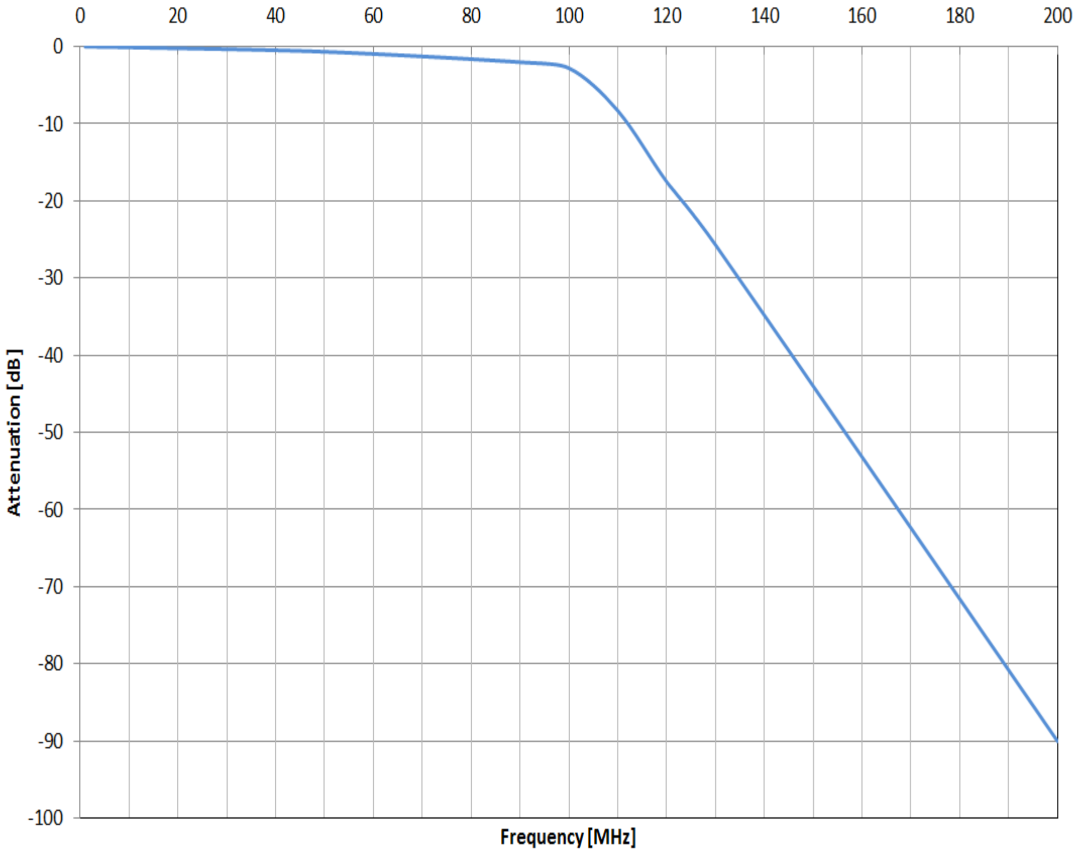


Figure 40: Frequency response of the anti-aliasing filter of WSMX HS

The maximum sample rate of the WSMX HS digitizer is 250 Msps. The usable bandwidth of a decimated waveform captured with this instrument depends on the selected sample rate. For sample rates equal or greater than 25 MHz it is 2/5 or 40% of the sample rate, which corresponds to 80% of the Nyquist frequency.

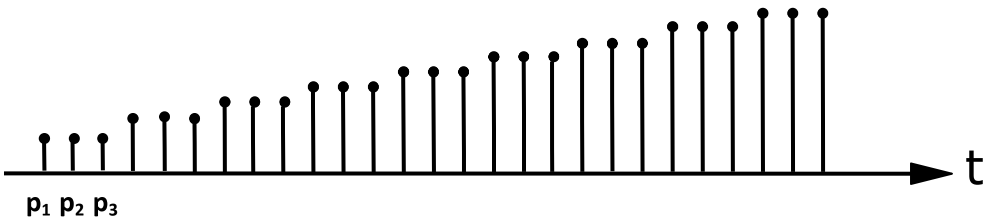
For sample rates lower than 25 MHz it is 1/5 or 20% of the sample rate, which corresponds to 40% of the Nyquist frequency. The minimum sample rate is 6.25 Msps without decimation, and 6.25 ksps with decimation. Without decimation, sampling rates lower than 6.25 Msps lead to a bind error.

4.4.9 Averaging

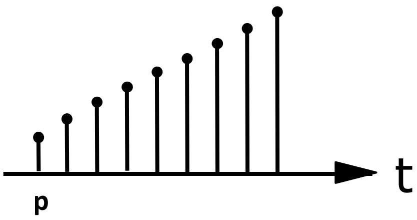
Signals are usually captured in the presence of significant noise; one important technique to improve the estimates of a value is averaging. WSMX digitizers offer two different kinds of averaging: averaging per sample and waveform averaging.

Averaging techniques remove noise from both the signal source and the measurement instrument; for this reason averaging techniques must not be used for noise measurements.

Averaging per sample
Averaging per sample improves dc value and linearity measurements. Each value is captured several times; these values are automatically summed up and divided by the number of repetitions.



$$p = \frac{1}{3}(p_1 + p_2 + p_3)$$

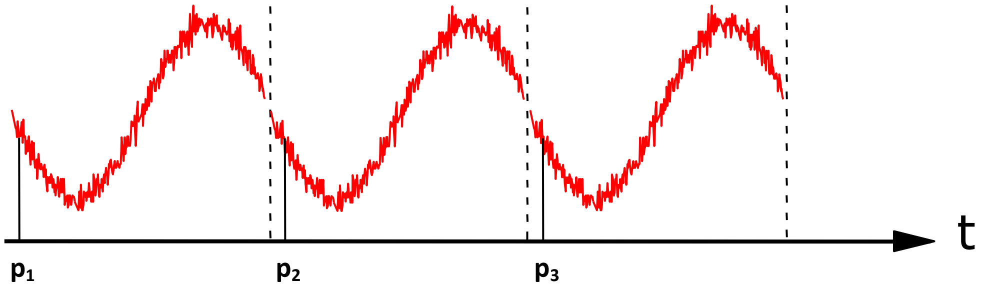


Use the action property *averagesPerSample* to average subsequent samples.

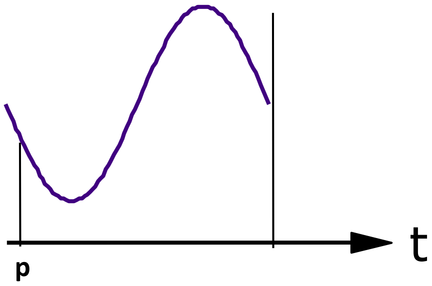
Example:
```
averagesPerSample = 8;
```

Waveform (Ensemble) Averaging

Waveform or ensemble averaging reduces noise in dynamic measurements. The captured vector is divided into n portions of equal length. Then the mean value of the corresponding data points is determined. Ensemble averaging reduces the noise of the signal source (DUT) and the digitizer; for this reason this method is not appropriate for noise measurements.



$$p = \frac{1}{3}(p_1 + p_2 + p_3)$$



Use the action property *waveformAveraging* to average entire waveforms.

Example:

```
waveformAveraging = 4;
```

4.4.10 FIR filter

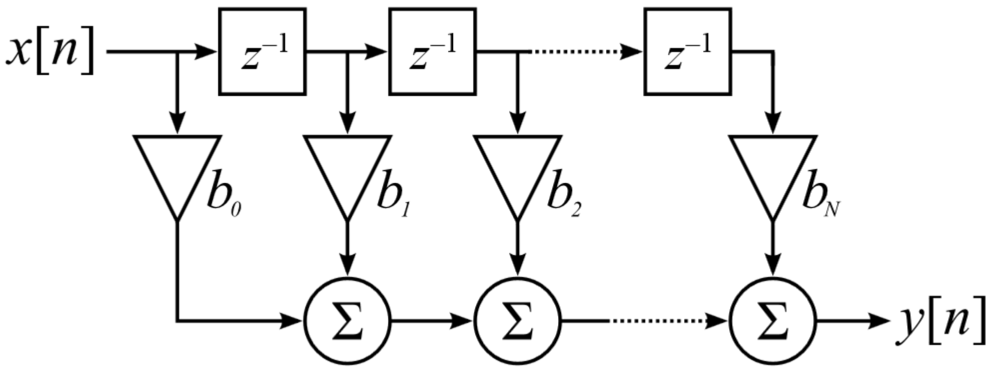


Figure 41: Structure of a FIR filter

The captured and digitized data can be filtered with a 16 to 128 tap, equiripple, multi-purpose FIR filter, either in the instrument's own signal processing unit (SPU) or in the workstation processor. This filter can be configured either as a standard low pass, high pass, band pass or band stop filter with a normalized gain (0 dB gain at the center frequency of the pass band). The filter can for example be used to reduce the bandwidth of the captured signal in order to achieve correlation with setups using legacy analog instruments.

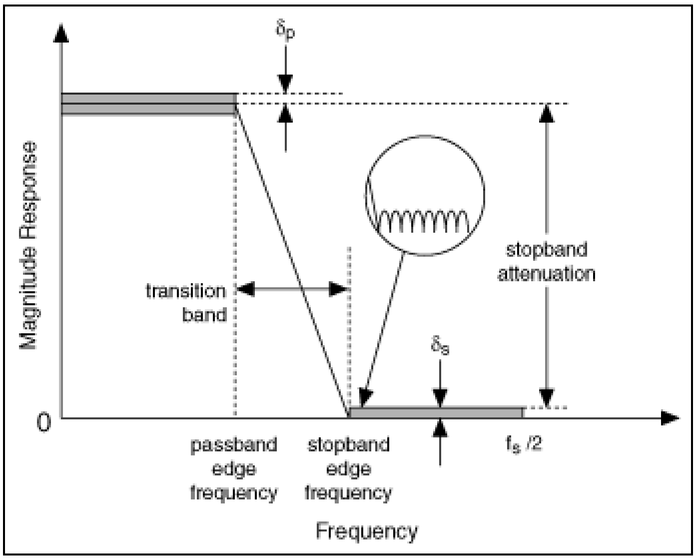


Figure 42: Filter characteristics

Use the instrument property group *digitalFilter* to set up the FIR filter.

1. Select the filter type: low, high or band pass, or band stop

```
digitalFilter.filterType  = lowPass;
```

2. Select the mode: hard- or software (optional)

```
digitalFilter.filterMode  = hardware;
```

3. Select either the cut-off frequency, or the passband and stopband edge frequencies

```
digitalFilter.cutOffFrequency1 = 15 MHz;  // cut-off freq in Hz

digitalFilter.passbandEdge = 14.0 MHz;    // passband edge in Hz
digitalFilter.stopbandEdge = 20.0 MHz;    // stopband edge in Hz
```

4. Select either the number of filter taps or the minimum roll-off. Note: the number of filter taps is ignored when the frequency response is determined by band edges (optional)

```
digitalFilter.numFilterTaps   = 128;    // 16–128 taps

// roll-off in dB per decade: 0.01-100 dB
digitalFilter.minRollOff   = 100 dB;
```

5. Select either the minimum stop band attenuation or the max. pass band ripple (optional)

```
// min. stopband attenuation in dB
digitalFilter.minAttenuation = 60 dB;

// max. passband ripple in dB
digitalFilter.maxRipple = 0.1dB;
```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The number of taps of a FIR Filter

The number of taps determines

- the flatness of the pass band
- the steepness of the roll-off
- the stopband attenuation

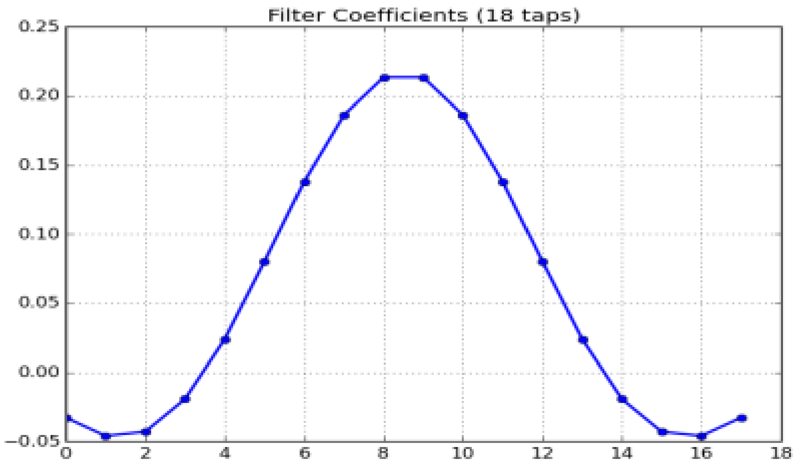

Figure 44: 18 tap FIR filter

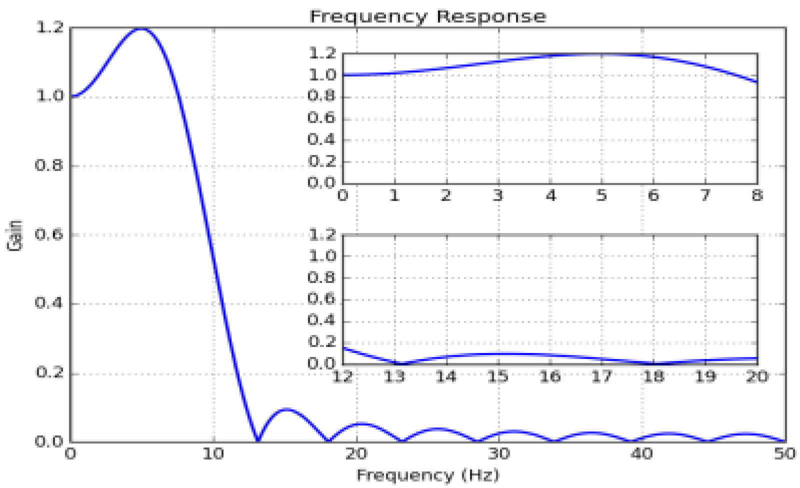

Figure 43: Frequency response of the 18 tap FIR filter

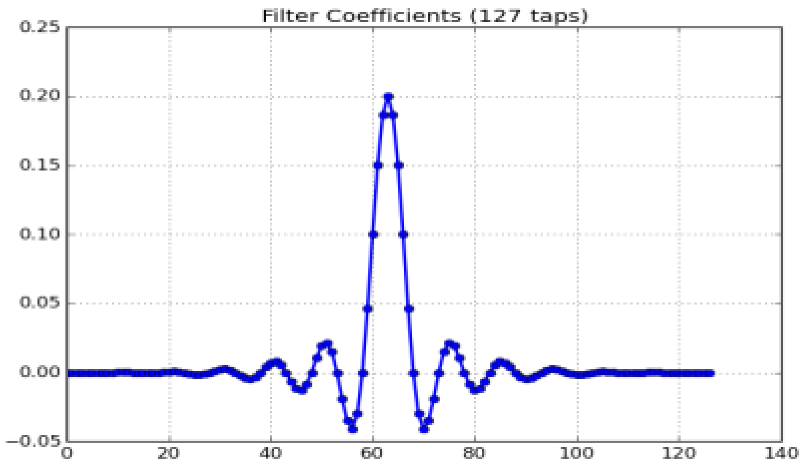

Figure 46: 127 tap FIR filter

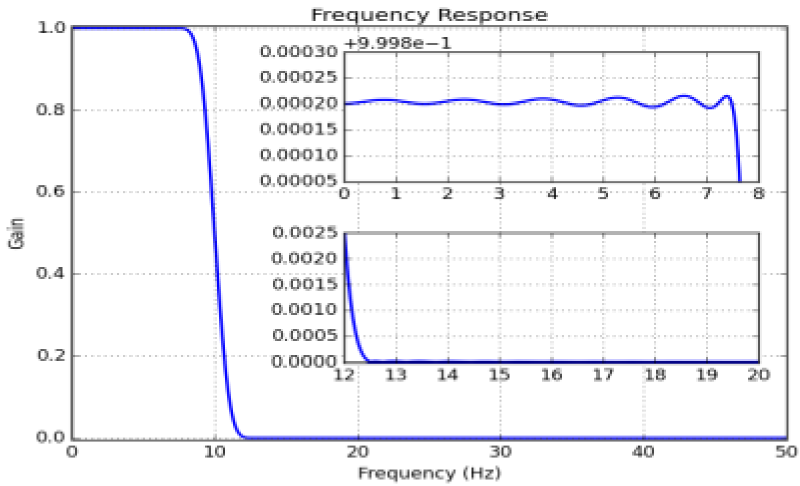

Figure 45: Frequency response of the 127 tap FIR filter

4.4.11 DownConversion

Digital down-conversion converts a digitized, real, band limited signal down to a real or complex, lower frequency signal. A numerical oscillator (NCO) generates a complex sinusoid at the analog center frequency. The input waveform is then multiplied with this sinusoid, resulting in a complex waveform, with images of the original waveform centered at the sum and difference frequency.

WSMX HS digitizer channels contain a built-in downconverter, which multiplies the captured samples with the output data of a numerical oscillator to split IF data into the in-phase (I) and the quadrature (Q) data, or simply to shift captured data into a different frequency band. This option is only available on Wave Scale high speed modules.

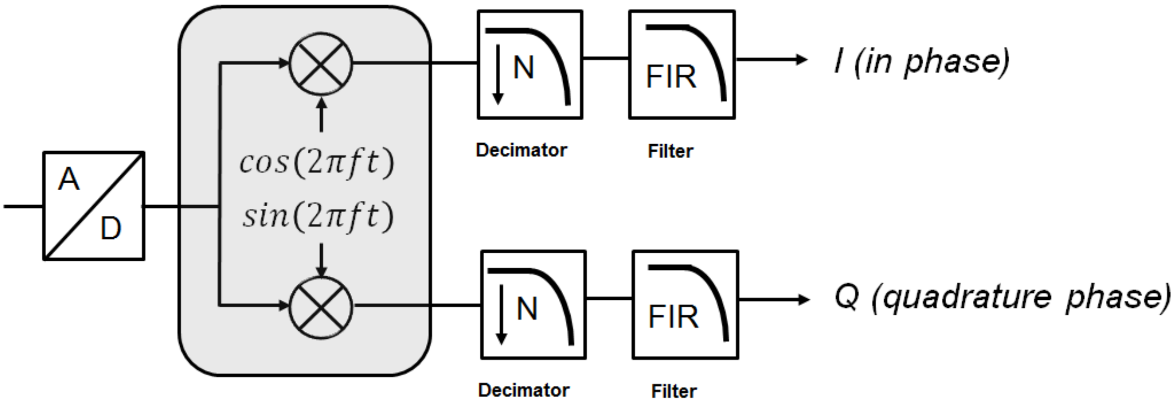

Figure 47: Complex downconversion with decimator and filter

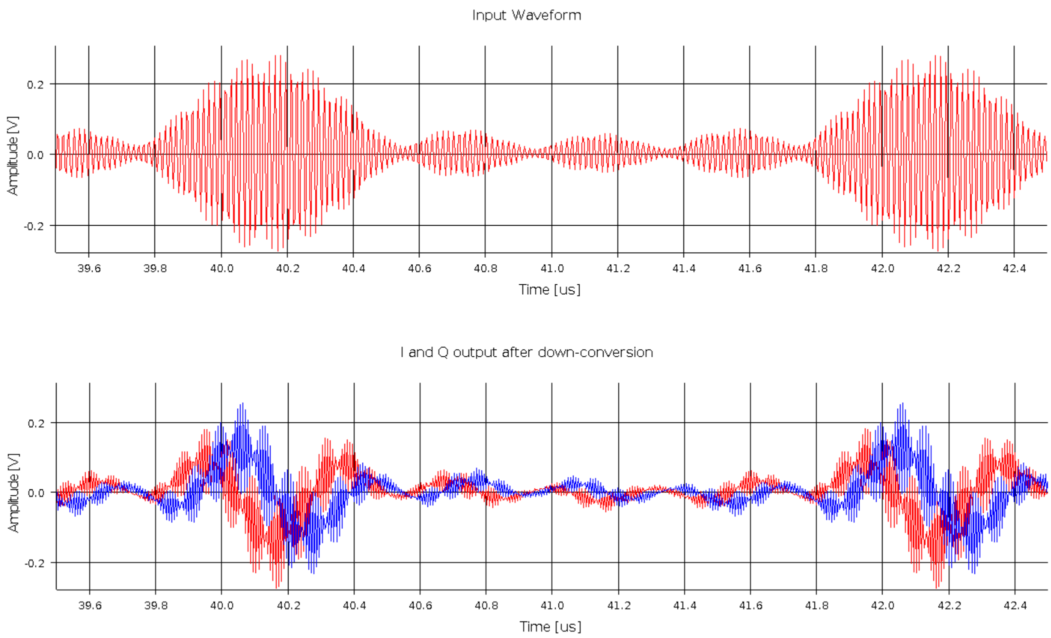

Figure 48: Signal in the range from 59 to 61 MHz, down-converted with an analog center frequency of 58 MHz

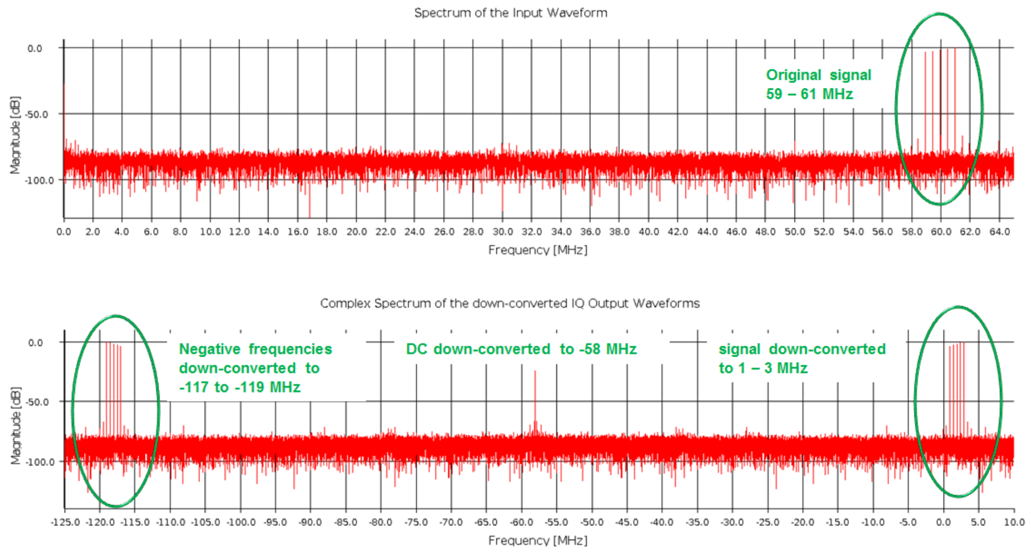

Figure 49: Spectra of the in- and output waveform

Use the instrument property group *downConversion* to set up the down-converter. *downConversion.analogCenterFrequency* sets the analog center frequency, which is the frequency of the sinusoids. If *downConversion.iqConversion* is set to *true*, the resulting output waveform is a complex IQ waveform, if set to *false* only the in-phase waveform is returned.

Example:

```
downConversion.analogCenterFrequency = 58 MHz;
downConversion.iqConversion          = true;
```

A low pass filter after the down-converter can pass the difference frequency and reject the sum frequency image. The resulting complex baseband representation of the original signal can be decimated to a lower sampling rate without losing any information contained in the original signal.

4.4.12 Target Frequency

Use the property *targetFrequency* to optimize the amplitude accuracy. This property is only relevant for WSMX HS modules and has no effect on WSMX HR modules; it is mandatory in case the *IQ* instrument option is set.

Default: no default value

Example:

```
targetFrequency = 5 MHz;
```

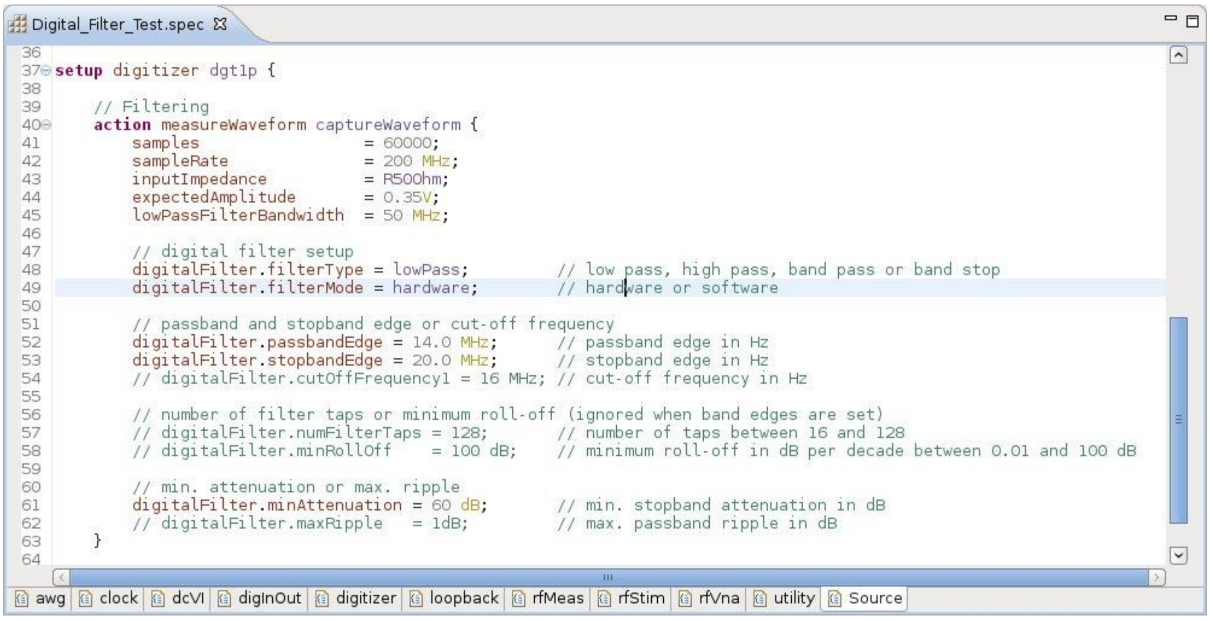

Figure 50: WSMX digitizer Setup

4.5 Digitizer Setup with the Device Setup API

In addition to the SmarTest Setup Format (SSF), SmarTest's own domain specific language, AWGs and digitizers can also be set up using the Device Setup API, the Java test method interface to the instruments. The Device Setup API provides a complete set of functions to set up measurements. The example below shows a complete digitizer setup.

```
package Setup_Examples;

import xoc.dsa.DeviceSetupFactory;
import xoc.dsa.IDeviceSetup;
import xoc.dsa.ISetupDigitizer;
import xoc.dta.TestMethod;
import xoc.dta.datatypes.dsp.MultiSiteSpectrum;
import xoc.dta.datatypes.dsp.MultiSiteWaveDouble;
import xoc.dta.datatypes.dsp.SpectrumUnit;
import xoc.dta.measurement.IMeasurement;
import xoc.dta.resultaccess.IDigitizerResults;
import xoc.dta.testdescriptor.IParametricTestDescriptor;

public class HeadphoneOutput extends TestMethod {

    public double signalFrequency = 997;
    public int    samples         = 2500;
    public double sampleRate      = 2500.0E3/51.0;

    public IMeasurement measurement;
    public IParametricTestDescriptor tdSNR_left, tdSND_left,
           tdTHD_left, tdAmplitude_left, tdOffset_left, tdSNR_right,
           tdSND_right, tdTHD_right, tdAmplitude_right,
           tdOffset_right;

    @Override
    public void setup(){

    IDeviceSetup ds = DeviceSetupFactory.createInstance();

    ///////////////////////////////////////////////////////////
    // Set up the digitizer instrument
    ///////////////////////////////////////////////////////////

    ISetupDigitizer digitizerSetup = ds.addDigitizer("HPR_P+HPL_P");
    digitizerSetup.setConfigOptionDifferential();

    ISetupDigitizer.IMeasureWaveform Capture_Waveform =
            digitizerSetup.measureWaveform("Capture_Waveform");
    Capture_Waveform.setSamples(50000);
    Capture_Waveform.setSampleRate(sampleRate);
    Capture_Waveform.setExpectedAmplitude(1.40);
    Capture_Waveform.setLowPassFilterBandwidth(signalFrequency);
    Capture_Waveform.setDecimation(true);
```

```
        ////////////////////////////////////////////////////
        // Generate the operating sequence
        ////////////////////////////////////////////////////

        ds.parallelBegin("ParallelGrp1");
        {
             ds.sequentialBegin("SequentialGrp1");
             {ds.actionCall(Capture_Waveform);}
             ds.sequentialEnd();
        }
        ds.parallelEnd();

     measurement.setSetups(ds);

     }


@Override
public void execute() {

     measurement.execute();

     IDigitizerResults results =
               measurement.digitizer("HPR_P+HPL_P").preserveResults();

     // (…)
     }

  }
```

5. Synchronization

5.1 Preparation times

All analog actions need to be prepared before they can be executed: signal paths and clock frequencies need to be switched, the registers of the converters need to be reprogrammed, digital filter parameters need to be loaded, etc. During these preparation times, the execution of actions is not possible.

These preparation times do neither include filter settling times, nor transient responses of the DUT or the DUT board. Take them into account by extending waveforms and removing samples which have been captured during transition times.

5.2 Delay

Actions can be delayed with the *delay* property, with the wait statement inside the operating sequence or by attaching the action to a different vector in the pattern.

5.3 Sync Accuracy

SmarTest synchronizes action starts as precise as possible for the given analog sample rates, digital waveform periods, and wait times. This can lead to long synchronization times or even to conflicts, which SmarTest cannot resolve.

Coherently captured, periodic analog waveforms usually do not require precise synchronization with other instruments, so that in many cases these conflicts can be resolved, or sync times can be shortened by relaxing the sync accuracy to a value in the range of tens or hundreds of nanoseconds.

syncAccuracy only relaxes the absolute synchronization accuracy; even with relaxed requirements the actions are started in a deterministic manner.

By default action starts are synchronized with a default accuracy of 1 ns.

6. Physical Interface

Two different pogo cables available:

Standard high density pogo cable **E9740P**

Figure 51: Standard high density pogo cable E9740P

Compatibility single density pogo cable **E9740PSD**

Figure 52: Compatibility single density pogo cable

6.1 The standard high density pogo cable E9740P

One HD pogo connector connects to 8 units:

- connector A to units 1 to 8
- connector B to units 9 to 16

Differential routing is identical to digital HD pogos. The HD pogo connector is incompatible to AV8, MBAV8 or MBAV8+.

On the mixed board E9740A the units 1 and 2 are high-speed units and units 3 and 4 are high resolution units.

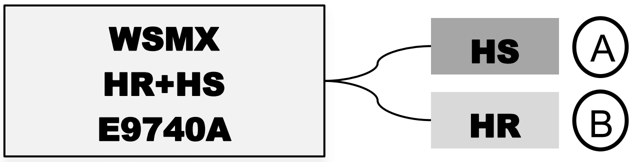

Figure 53: Pogo connector assignment of HD pogo cable

Model File

```
CARD_CONFIGURATION

# WSMX HR with HD pogo cable
slot = 101, HW = WSMX, pogoblocks=HR:(231 631 232 632), type=HD2x32
# WSMX HS with HD pogo cable
slot = 101, HW = WSMX, pogoblocks=HS:(231 631 232 632), type=HD2x32
# WSMX HS/HR with HD pogo cable
slot = 101, HW = WSMX, pogoblocks=HS:(231 631) HR:(232 632), type=HD2x32
```

HD Pogo connector A

The graph below shows the HD pogo connector A with the connected units and connectors.

1X1 = Unit 1, pogo X1, ■ = ground , o = orientation marker

	a	b	c	d
01	■	1X1	■	**o**
02	5X1	■	1X3	■
03	■	1X2	■	5X3
04	5X2	■	1X4	■
05	■	2X1	■	5X4
06	6X1	■	2X3	■
07	■	2X2	■	6X3
08	6X2	■	2X4	■
09	■	3X1	■	6X4
10	7X1	■	3X3	■
11	■	3X2	■	7X3
12	7X2	■	3X4	■
13	■	4X1	■	7X4
14	8X1	■	4X3	■
15	■	4X2	■	8X3
16	8X2	■	4X4	■
17	■	8X4	■	**o**

Figure 54: Connections of HD pogo connector A

legend: 1X1 = Unit 1, pogo X1, ■ = ground , o = orientation marker

Two logical pogo block numbers are associated with each physical HD pogo connector. The first one is used for the pogo rows b and c, the second one, whose number equals the first one plus 400, is used for the rows a and d. The graph below shows the SD pogo connector A with the pogo numbers.

nnn : 1st pogo block number
mmm: 2nd pogo block number = nnn + 400

Example:
nnn = 231
mmm = 631

Standard pogo locations:
1. HS board : 426 / 826
2. HS board : 320 / 720
3. HS board : 232 / 632
4. HS board : 319 / 719
1. HR board : 323 / 723
2. HR board : 432 / 832
3. HR board : 324 / 724
4. HR board : 427 / 827

	a	b	c	d
01	■	nnn01	■	o
02	mmm01	■	nnn02	■
03	■	nnn03	■	mmm02
04	mmm03	■	nnn04	■
05	■	nnn05	■	mmm04
06	mmm05	■	nnn06	■
07	■	nnn07	■	mmm06
08	mmm07	■	nnn08	■
09	■	nnn09	■	mmm08
10	mmm09	■	nnn10	■
11	■	nnn11	■	mmm10
12	mmm11	■	nnn12	■
13	■	nnn13	■	mmm12
14	mmm13	■	nnn14	■
15	■	nnn15	■	mmm14
16	mmm15	■	nnn16	■
17	■	mmm16	■	o

Figure 55: Pogo numbers of HD pogo connector A

The “native” connectors X1 and X2 for the AWG, and the connectors X3 and X4 for the digitizer provide better signal quality due to direct routing.

HD Pogo connector B

The graph below shows the HD pogo connector B with the connected units and connectors.

1X1 = Unit 1, pogo X1, ■ = ground , o = orientation marker

	a	b	c	d
01	■	9X1	■	o
02	13X1	■	9X3	■
03	■	9X2	■	13X3
04	13X2	■	9X4	■
05	■	10X1	■	13X4
06	14X1	■	10X3	■
07	■	10X2	■	14X3
08	14X2	■	10X4	■
09	■	11X1	■	14X4
10	15X1	■	11X3	■
11	■	11X2	■	15X3
12	15X2	■	11X4	■
13	■	12X1	■	15X4
14	16X1	■	12X3	■
15	■	12X2	■	16X3
16	16X2	■	12X4	■
17	■	16X4	■	o

Figure 56: Connections of HD pogo connector B

Two logical pogo block numbers are associated with each physical HD pogo connector. The first one is used for the pogo rows b and c, the second one, whose number equals the first one plus 400, is used for the rows a and d. The graph below shows the SD pogo connector B with the pogo numbers.

kkk: 3rd pogo block number
jjj: 4th pogo block number = kkk+ 400

Example:
kkk = 232
jjj = 632

	a	b	c	d
01	■	kkk01	■	**o**
02	jjj01	■	kkk02	■
03	■	kkk03	■	jjj02
04	jjj03	■	kkk04	■
05	■	kkk05	■	jjj04
06	jjj05	■	kkk06	■
07	■	kkk07	■	jjj06
08	jjj07	■	kkk08	■
09	■	kkk09	■	jjj08
10	jjj09	■	kkk10	■
11	■	kkk11	■	jjj10
12	jjj11	■	kkk12	■
13	■	kkk13	■	jjj12
14	jjj13	■	kkk14	■
15	■	kkk15	■	jjj14
16	jjj15	■	kkk16	■
17	■	jjj16	■	**o**

Figure 57: Pogo numbers of HD pogo connector B

6.2 The compatibility single density pogo cable E9740PSD

A compatibility pogo cable provides loadboard and pogo compatibility for devices that use **either** high-speed **or** high-resolution cores. One Wave Scale MX board can replace two AV8, MBAV8 or MBAV8+ boards. The SD compatibility pogo cable is compatible to AV8, MBAV8 and MBAV8+.

One pogo connector connects to four units:

- connector A to units 1 to 4,
- connector B to units 5 to 8
- connector C to units 9 to 12
- connector D to units 13 to 16

On the mixed board E9740A the units 1 to 8 are high-speed units and units 9 to 16 are high resolution units.

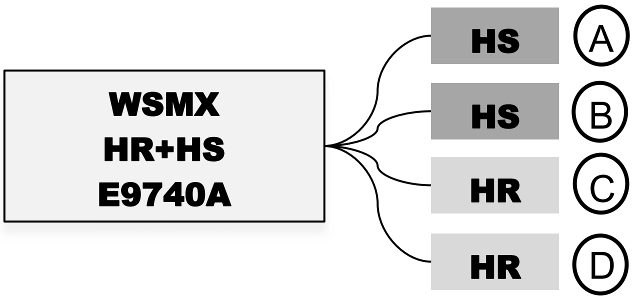

Figure 58: Pogo connector assignment of SD pogo cable

Model File

```
CARD_CONFIGURATION

# WSMX HR with SD pogo cable
slot = 101, HW = WSMX, pogoblocks=HR:(231 431 232 432)
# WSMX HS with SD pogo cable
slot = 101, HW = WSMX, pogoblocks=HS:(231 431 232 432)
# WSMX HS/HR with SD pogo cable
slot = 101, HW = WSMX, pogoblocks=HS:(231 431) HR:(232 432)
```

The graphs below show the SD pogo connectors A, B, C and D with the connected units and connectors.

Pogo connector A

	a	b	c	d
01	■	4X4	■	o
02		■	4X3	■
03	■	2X4	■	
04		■	2X3	■
05	■	3X4	■	
06		■	3X3	■
07	■	1X4	■	
08		■	1X3	■
09	■	4X2	■	
10		■	4X1	■
11	■	2X2	■	
12		■	2X1	■
13	■	3X2	■	
14		■	3X1	■
15	■	1X2	■	
16		■	1X1	■
17	■	n/c	■	o

Pogo connector B

a	b	c	d
■	8X4	■	o
	■	8X3	■
■	6X4	■	
	■	6X3	■
■	7X4	■	
	■	7X3	■
■	5X4	■	
	■	5X3	■
■	8X2	■	
	■	8X1	■
■	6X2	■	
	■	6X1	■
■	7X2	■	
	■	7X1	■
■	5X2	■	
	■	5X1	■
■	n/c	■	o

Figure 59: Connections of SD pogo connectors A and B

Pogo connector C **Pogo connector D**

	a	b	c	d		a	b	c	d
01	■	12X4	■	**o**		■	16X4	■	**o**
02		■	12X3	■			■	16X3	■
03	■	10X4	■			■	14X4	■	
04		■	10X3	■			■	14X3	■
05	■	11X4	■			■	15X4	■	
06		■	11X3	■			■	15X3	■
07	■	9X4	■			■	13X4	■	
08		■	9X3	■			■	13X3	■
09	■	12X2	■			■	16X2	■	
10		■	12X1	■			■	16X1	■
11	■	10X2	■			■	14X2	■	
12		■	10X1	■			■	14X1	■
13	■	11X2	■			■	15X2	■	
14		■	11X1	■			■	15X1	■
15	■	9X2	■			■	13X2	■	
16		■	9X1	■			■	13X1	■
17	■	n/c	■	**o**		■	n/c	■	**o**

Figure 60: Connections of SD pogo connectors C and D

SD pogo naming

SD pogo connectors only use the pogo rows b and c for the signals. One logical pogo block number is associated with each physical SD pogo connector. The graph below shows the SD pogo connector A with the pogo numbers.

nnn : pogo block number

Example:
nnn = 231 / 232 /431 / 432

	a	b	c	d
01	■	nnn01	■	o
02		■	nnn02	■
03	■	nnn03	■	
04		■	nnn04	■
05	■	nnn05	■	
06		■	nnn06	■
07	■	nnn07	■	
08		■	nnn08	■
09	■	nnn09	■	
10		■	nnn10	■
11	■	nnn11	■	
12		■	nnn12	■
13	■	nnn13	■	
14		■	nnn14	■
15	■	nnn15	■	
16		■	nnn16	■
17	■		■	o

Figure 61: Pogo numbers of SD pogo connectors

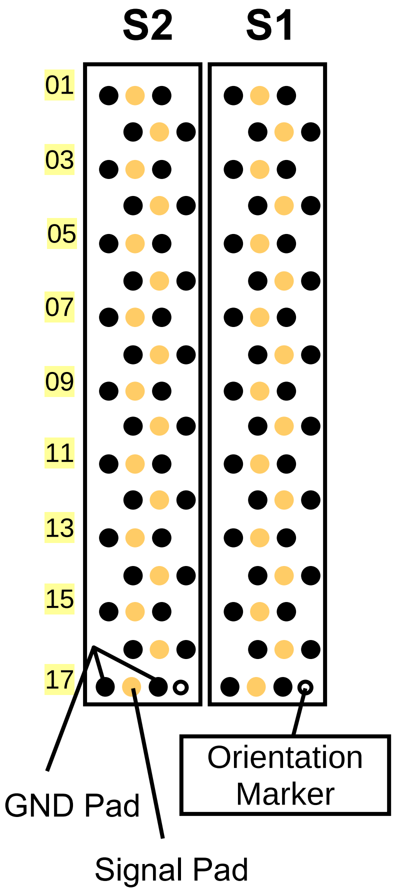

Figure 62: MBAV8 pogo connector

Pogo Pad	AV8 / MBAV8 S2 pogo Port	Unit	Wave Scale MX SD pogo B Port	AV8 / MBAV8 S1 pogo Port	Unit	Wave Scale MX SD pogo A Port
01	DD-	4	8X4	HH-	8	4X4
02	DD+	4	8X3	HH+	8	4X3
03	CC-	3	6X4	GG-	7	2X4
04	CC+	3	6X3	GG+	7	2X3
05	BB-	2	7X4	FF-	6	3X4
06	BB+	2	7X3	FF+	6	3X3
07	AA-	1	5X4	EE-	5	1X4
08	AA+	1	5X3	EE+	5	1X3
09	D-	4	8X2	H-	8	4X2
10	D+	4	8X1	H+	8	4X1
11	C-	3	6X2	G-	7	2X2
12	C+	3	6X1	G+	7	2X1
13	B-	2	7X2	F-	6	3X2
14	B+	2	7X1	F+	6	3X1
15	A-	1	5X2	E-	5	1X2
16	A+	1	5X1	E+	5	1X1
17	CT1		n/c	CT2		n/c

Figure 63: SD pogo compatibility to AV8, MBAV8 and MBAV8+

Pogo connectors C & D connect WSMX units 9-16

7. Calibration and Diagnostics

7.1 WSMX calibration and diagnostics activities

The mixed signal calibration and diagnostic are started from the control desk of the SmarTest8 maintenance perspective, a dedicated Eclipse perspective which includes an overall maintenance status view.

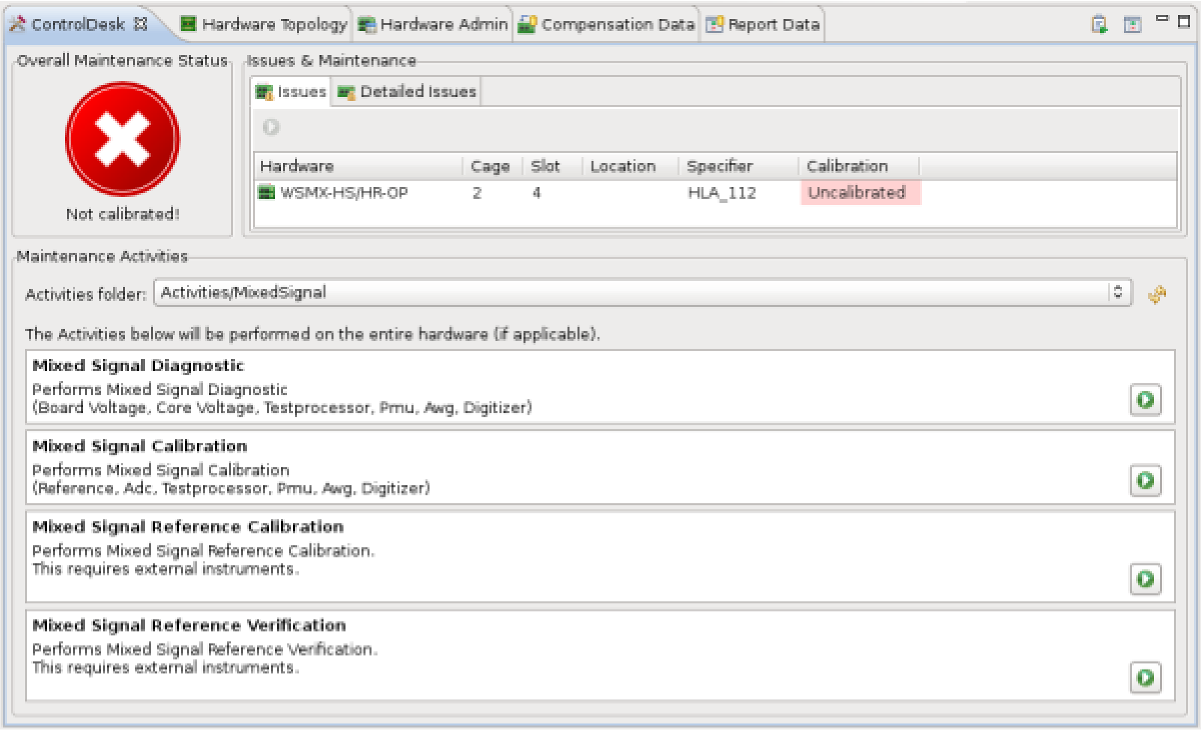

Figure 64: Maintenance Control Desk

Four mixed signal maintenance activities can be selected:

- Mixed Signal Reference Calibration (former *Base Cal*, first part of former *NIST Cal*)
- Mixed Signal Reference Verification (former *Trace Cal*, second part of former *NIST Cal*)
- Mixed Signal Reference Calibration
- Mixed Signal Reference Diagnostic

7.2 Mixed Signal Calibration

The mixed signal calibration includes timing and DC calibration and routines which are essential for the board to work properly (e. g. internal data interfaces). Standard calibration does not require external equipment; during calibration no board must be docked. The basic maintenance period (diagnostic interval) is 3 months.

The calibration of subsets is possible and the system software knows the dependencies. Calibration data are stored in a data base. The user can switch between calibration data sets. IQ calibration is fully integrated.

The following items are calibrated:

- Reference
- Testprocessor timing and interfaces
- PMU
- AWG & digitizer offset and DC level
- IQ amplitude and phase mismatch

7.3 WSMX I/Q phase & gain balance calibration

Different boards are required for HD and SD pogo cable wiring and for AWG and digitizer calibration.

	WSMX HS AWG **IQ Source Cal**	**WSMX HS digitizer** **IQ Meas Cal**
SD Pogo Connector	E9728B SRC IQ Calibration Board	E9728A MEAS IQ Calibration Board
HD pogo connector	E7010F Diagnostic Interconnection Board	E9740IQ IQ Calibration Board

Figure 65: Calibration Boards

7.4 Diagnostics

Diagnostics detects individual defective units, which could be disabled if unused. No external equipment is required for standard WSMX diagnostics. Full system diagnostics including digital, DC, Mixed-Signal and RF can be done without human interaction. No board must be docked during the diagnostic run.

8. Licenses

Base Capabilities & Licenses

Each Wave Scale MX board *E9740HS* includes floating licenses N9740HSBC for 4 units as base capability. Additional units can be enabled by additional, floating licenses *N5941UP.* The base license enables the AWG and digitizer with bandwidths to 30 MHz and the HS digitizer sample rate to 100 Msps. The speed upgrade license *N5942UP* enables the full bandwidths and sample rates. The license *N5940AF* enables all features.

All licenses are available as 30 day licenses (30DA), annual licenses (-AS) and perpetual licenses (-PRM).

Part Number	Description
N9740HSBC-PRM	Wave Scale MX high-speed base capability license, perpetual
N9740HSBC-AS	Wave Scale MX high-speed base capability license, annual subscription
N9740HSBC30DA	Wave Scale MX high-speed base capability license, 30 days
N9740HRBC-PRM	Wave Scale MX high-resolution base capability license, perpetual
N9740HRBC-AS	Wave Scale MX high-resolution base capability license, annual subscription
N9740HRBC30DA	Wave Scale MX high-resolution base capability license, 30 days
N5940UPRM	Wave Scale MX high-resolution unit license, perpetual
N5940UP-AS	Wave Scale MX high-resolution unit license, annual subscription
N5940UP30DA	Wave Scale MX high-resolution unit license, 30 days
N5941UPRM	Wave Scale MX high-speed unit license, perpetual
N5941UP-AS	Wave Scale MX high-speed unit license, annual subscription
N5941UP30DA	Wave Scale MX high-speed unit license, 30 days
N5942UPRM	Wave Scale MX high-speed bandwidth upgrade license, perpetual
N5942UP-AS	Wave Scale MX high-speed bandwidth upgrade license, annual subscription
N5942UP30DA	Wave Scale MX high-speed bandwidth upgrade license, 30 days
N5940AF12M-NL	Wave Scale MX all features, 12 months per board node-locked demo license
N5940AF30DA-NL	Wave Scale MX all features, 30 days per board node-locked demo license

Table 5: WSMX licenses

9. Performance Optimization

9.1 General recommendations

9.1.1 Native Connectors

The performance of an AWG or digitizer depends on the selected pogo connector. The connectors X1 and X2 are directly connected to AWG output, and the connectors X3 and X4 are directly connected to the digitizer input. Using these connectors usually yields better performance than the crossed connections. On WSMX HR the AWG can be connected to X1 and X2, and the digitizer can be connected to X3 and X4, however not at the same time.

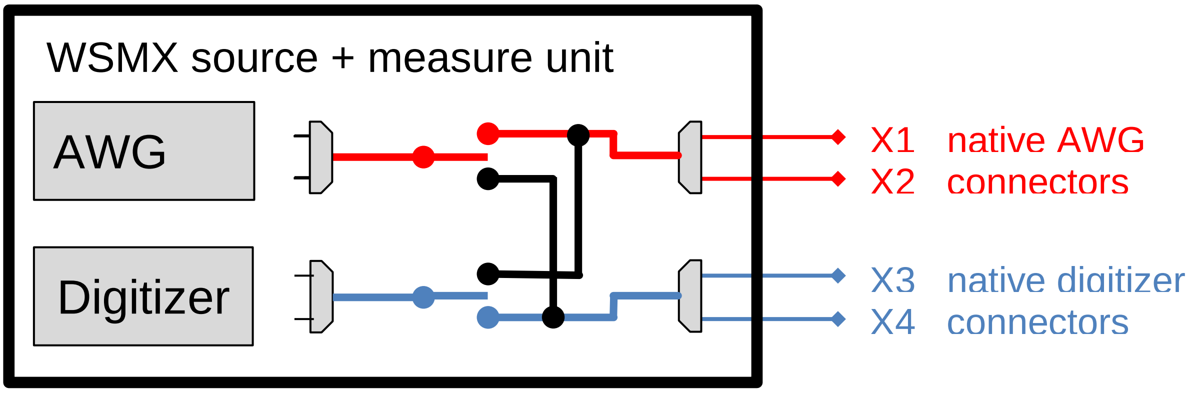

Figure 66: Native AWG and digitizer connectors

9.1.2 Standard Waveforms

The signal quality can differ significantly between setups using standard and sample-based waveforms. When standard waveforms are used in the setup, SmarTest optimizes the instrument setup is optimized for maximum performance (sample rate, filter selection, compensation & pre-distortion). The Wave Scale MX high resolution AWG even contains different D/A converters; some are optimized for static and others for dynamic signals. Only when standard waveforms are used in the setup, SmarTest selects the most suitable resource.

9.2 WSMX HR AWG

9.2.1 Settling times

The analog and digital filters of the AWG have long settling times. It is recommended to start the AWG up to 10 ms before the DUT starts digitizing. The settling times of the analog filters are as follows:

1 kHz filter: 7 ms
20 kHz filter: 500 us
80 kHz filter: 300 us

9.2.2 Distortion

The distortion varies significantly with the amplitude range setting, the selected sampling rate and the amplitude of the signal. Distortion can be reduced by using the largest amplitude range and a low sampling rate. The best distortion performance can be achieved for amplitudes around 1 Vrms.

The distortion performance differs between the positive and the negative leg of a differential pin pair: the positive shows better performance for signal amplitudes lower than 2.4 V_{rms} and the negative leg performs better at signal amplitudes higher than 2.4 V_{rms}.

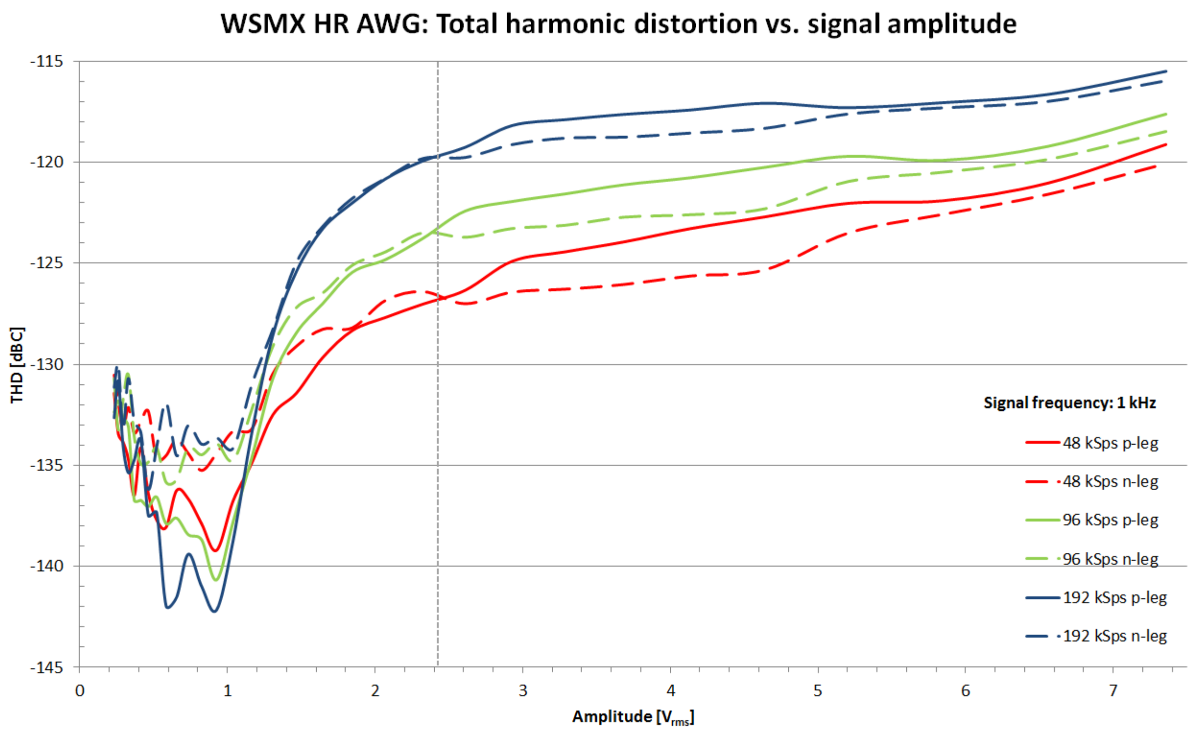

Figure 67: THD vs. signal amplitude of the WSMX HR AWG

9.3 WSMX HR Digitizer

We recommend differential connections or the ground sense option, as these connection modes compensate for many sources of errors, and thus usually yield more accurate results.

9.4 WSMX HS AWG

Different offsets on the two legs of a differential pair lead to heavily increased harmonics.

9.5 WSMX HS Digitizer

9.5.1 Input voltage Range

The programmable gain amplifier of the digitizer input allows for dynamic ranging with voltage ranges from 62.5 mV_{peak} to 2 V_{peak}. The system software selects the appropriate input range based on the setup property *expectedAmplitude*.

The dynamic performance of the digitizer is range-dependent, and also the harmonic distortion, but to a smaller degree. For this reason it is recommended to set the *expectedAmplitude* to a value close the maximum expected amplitude without much of a margin. The automatic range selection already adds a margin to the *expectedAmplitude.*

The figures below show the noise density, the THD and the SNR for the different ranges.

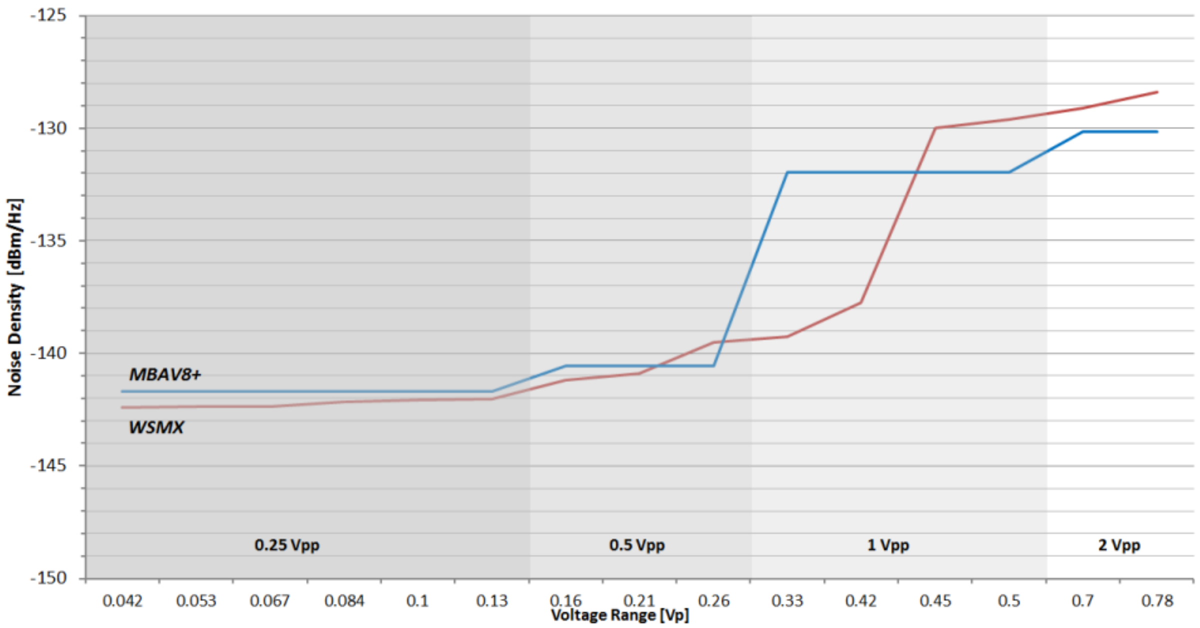

Figure 68: Noise density of the WSMX HS digitizer (red)

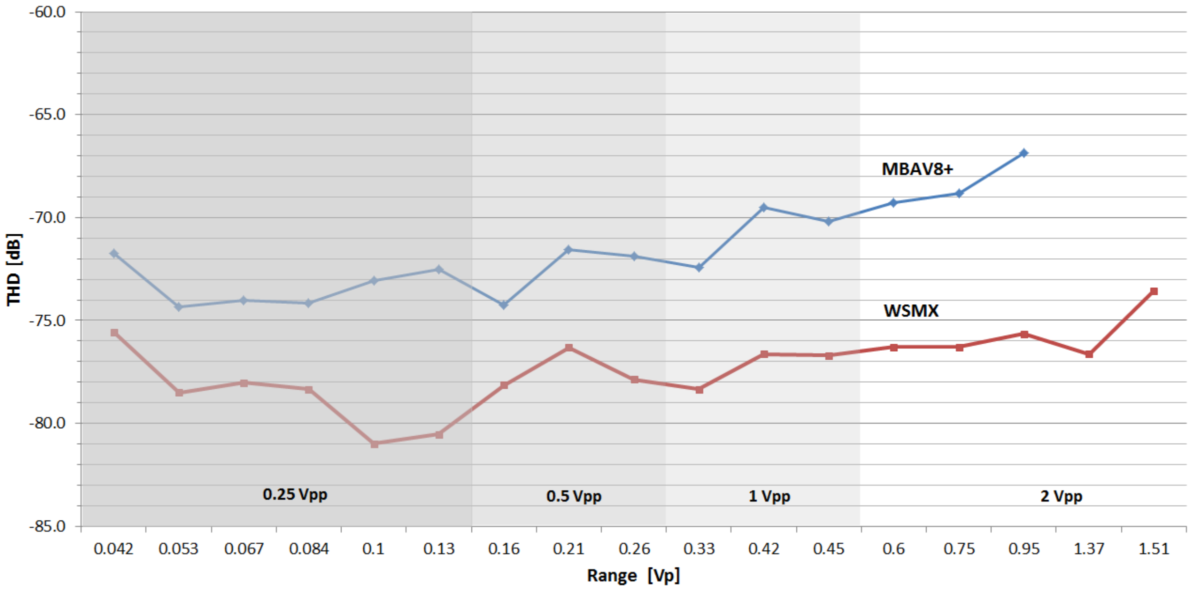

Figure 69: Harmonic distortion of the WSMX HS digitizer (red)

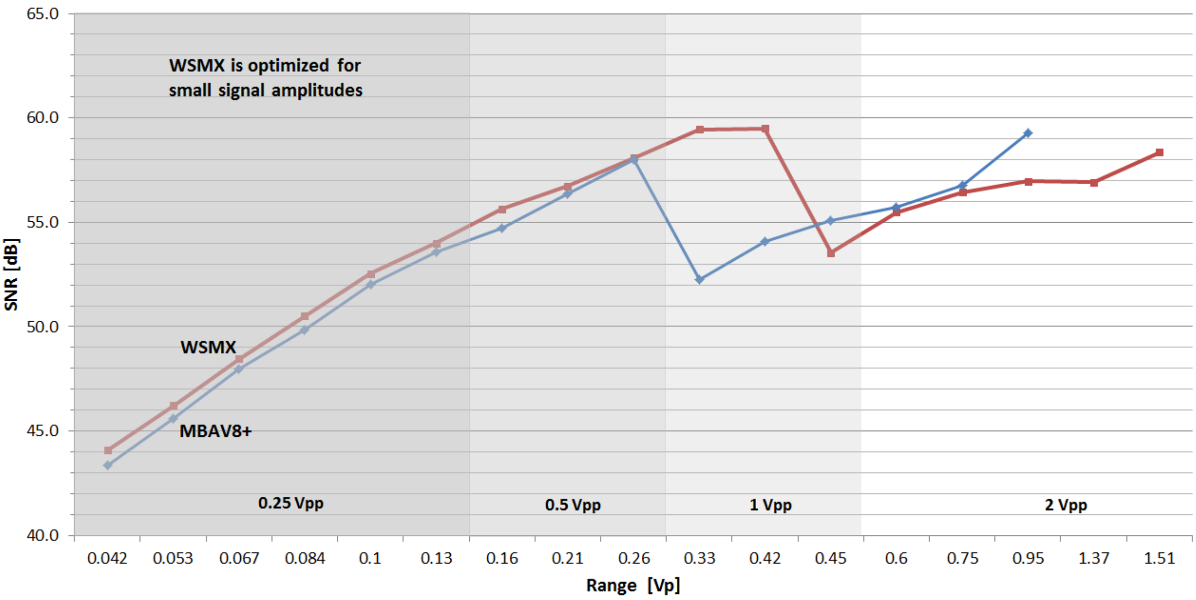

Figure 70: Signal-to-Noise ratio of the WSMX HS digitizer (red)

10. Performance Comparison

10.1 WSMX HS/MBAV8+ VHF AWG

	WSMX – HS AWG	**MBAV8+ - VHF AWG**
Amplitude ranges	40 mV_{pp} - 2.5 V_{pp} into 50 Ω	2.5 V_{pp} into 50 Ω
Offset ranges	-2 V to 4 V into open	-1.5 V to 2.5 V into 50 Ω
Sample rates	10 - 500 Msps	8 ksps - 500 Msps
Max. sine wave frequency (-3 dB frequency)	220 MHz	200 MHz
DC accuracy (into open)	± (0.4% of setting + 8 mV + 0.2% of DC offset)	±(1.3% of setting + 35 mV)
Memory	216 Ms (shared between 2 AWG + 2 digitizers)	31.7 Ms per source unit
THD	-81 dBc (10 MHz signal, 0 dBm, 50 Ω) -55 dBc (100 MHz signal, 0 dBm, 50 Ω)	-75 dBc (10 MHz signal, 0 dBm, 50 Ω) -53 dBc (100 MHz signal, 0 dBm, 50 Ω)
SFDR	81 dBc (10 MHz signal, 0 dBm, 50 Ω)	75 dBc (10 MHz signal, 0 dBm, 50 Ω)
Output impedance	50 Ohm ± 5%	50 Ohm ± 5%
Analog reconstruction filters (0.1 dB / 3dB)	5 / 5.7 MHz, 30 / 37MHz, 80 / 88 MHz, 200 / 220 MHz	58 MHz, 200 MHz
Noise density (typical)	-141 dBm/Hz @ 10 MHz for signal amplitudes <1.75 V_{pp} into open	

Table 6: Performance comparison WSMX HS vs. MBAV8+ VHF AWG

10.2 WSMX HS/MBAV8+ VHF Digitizer

	WSMX – HS digitizer	MBAV8+ - VHF digitizer
Amplitude ranges	62.5 mV_{pp} to 3 V_{pp}, 2dB steps. Range selection based on setup property *expectedAmplitude*.	±0.125 V to ±2 V
Offset ranges	-2.5 V to 2.5 V	-3 V to 3 V
Sample rates	6.25 to 250 Msps	1 to 180 Msps
DC accuracy	±(0.75% of reading + 0.3% of offset + 5 mV)	±0.5% of reading ±0.3% of offset ± 5 mV
Memory	216 Ms (shared between 2 AWG + 2 digitizers)	14 Ms per measure unit
Bandwidth	300 MHz	300 MHz
THD	-79 dBc (10 MHz, 250 mVpk, 50 Ohms) -74 dBc (50 MHz, 250 mVpk, 50 Ohms)	-75 dBc (10 MHz, 250 mVpk, 50 Ohms) -68 dBc (50 MHz, 250 mVpk, 50 Ohms)
Input impedance	single-ended: 50 Ohm ± 5%, 100 kOhm ± 10% differential: 100 Ohm ± 5%	single-ended: 1 MΩ, 10 kΩ, 50 Ω, 37.5 Ω differential: 2 MΩ, 20 kΩ, 100 Ω, 75 Ω
Analog Filters (0.1 / 3dB)	100 / 107 MHz 300 / 340 MHz	15 MHz, 50 MHz, 105 MHz and 300 MHz
Noise density (typical)	142 dBm/Hz for 100 mV_p, 141 dBm/Hz for 200 mV_p 139 dBm/Hz for 300 mV_p, 130 dBm/Hz for 500 mV_p	

Table 7: Performance comparison WSMX HS vs. MBAV8+ VHF digitizer

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